

VitaSpec_huiles

ME

2024-05-10

Huiles achetées dans différents pays d'Afrique de l'Ouest en 2024. Analysées par Amel Joualli

Dans les spectres, manquent les échantillons 34 et 87

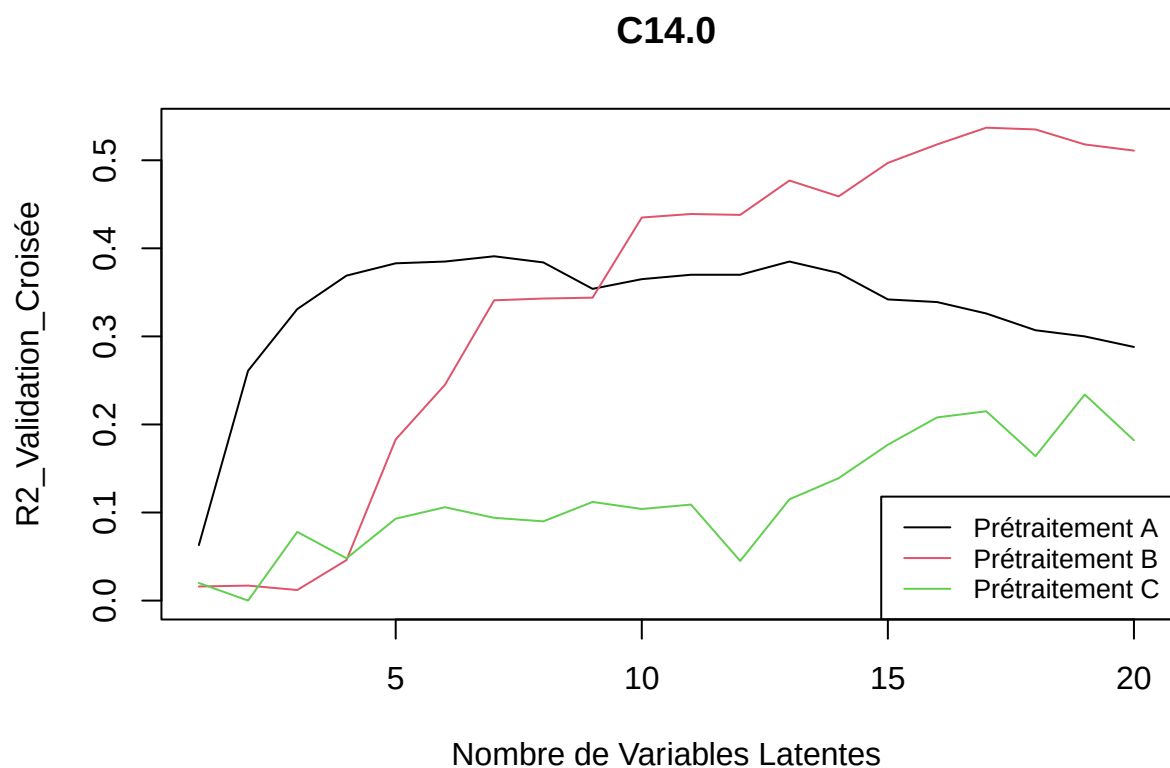
Mise en forme des données et save

Chargement des prétraitements

Calibrations sur spectres moyen

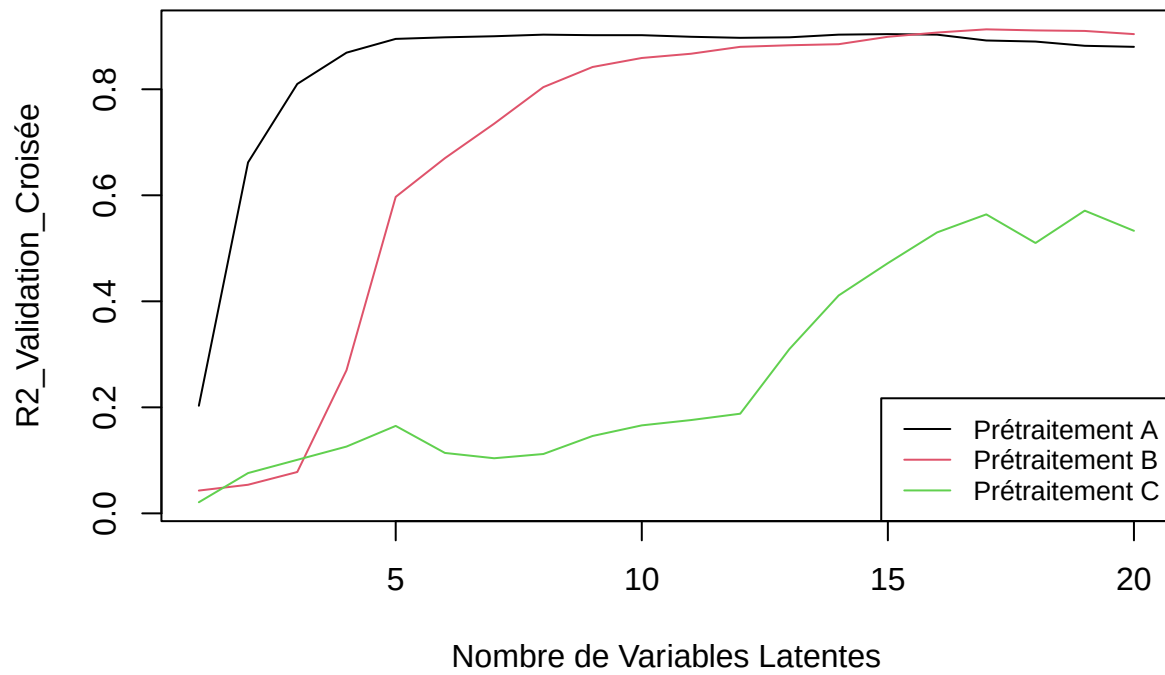
Prétraitement A: "1300-1750nm + SNV + DERIV.2nde"; Prétraitement B: "350-2500 nm + SNV + DERIV.2nde"; Prétraitement C: "350-1200nm + SNV"

```
## [1] "C14.0"
```

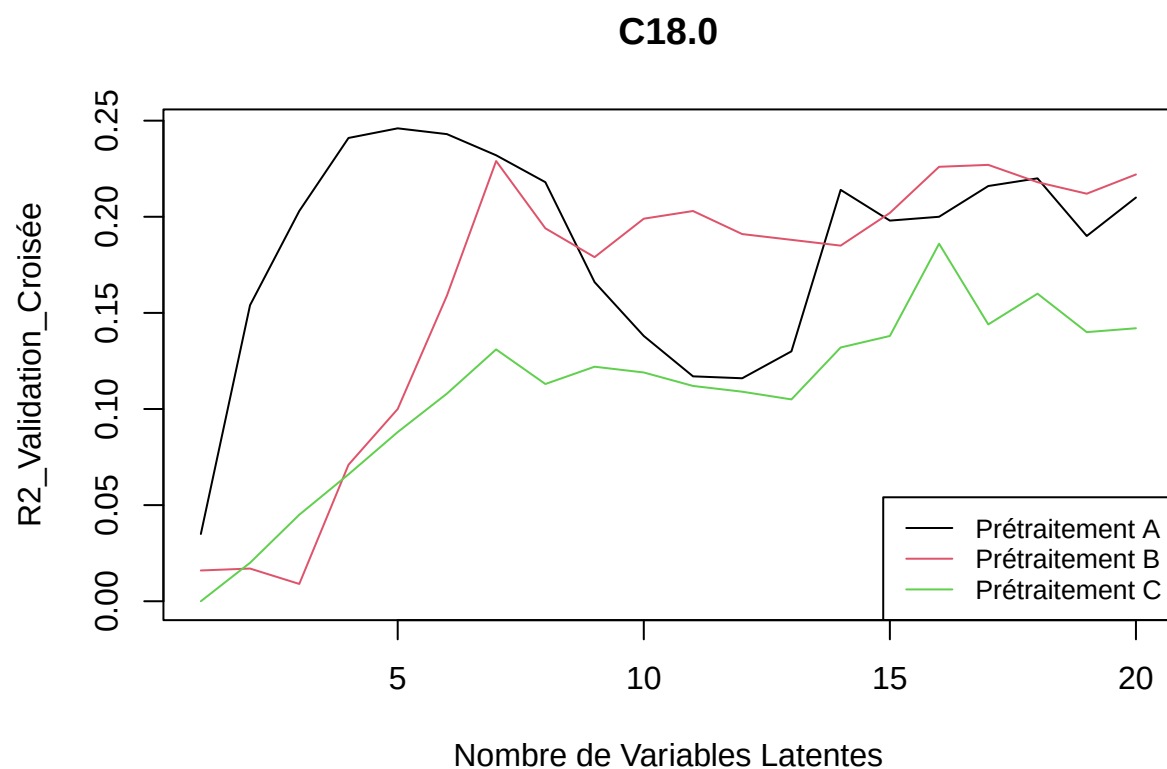


```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.016 0.017 0.012 0.046 0.183 0.245 0.341 0.343 0.344 0.435 0.439
## [13] 0.438 0.477 0.459 0.497 0.518 0.537 0.535 0.518 0.511
## [1] "SEP"
## [1] 0.211 0.213 0.220 0.211 0.211 0.192 0.185 0.173 0.173 0.174 0.163 0.162
## [13] 0.163 0.158 0.160 0.154 0.150 0.148 0.149 0.154 0.156
##
##
## [1] "C16.0"
```

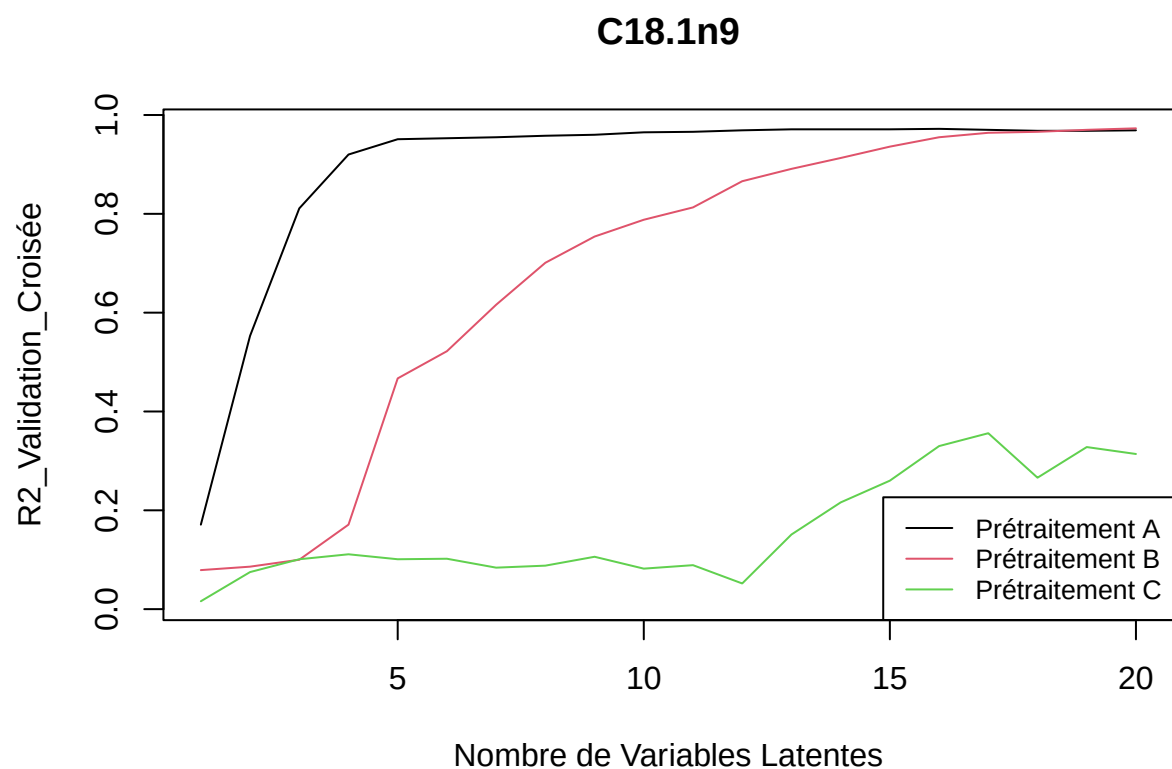
C16.0



```
## [1] "Prétraitement A"
## [1] "R2"
## [1] 1.000 0.203 0.662 0.810 0.869 0.895 0.898 0.900 0.903 0.902 0.902 0.899
## [13] 0.897 0.898 0.903 0.904 0.903 0.892 0.890 0.882 0.880
## [1] "SEP"
## [1] 2.781 2.467 1.603 1.202 0.999 0.894 0.879 0.871 0.860 0.864 0.865 0.880
## [13] 0.887 0.883 0.863 0.856 0.866 0.915 0.923 0.958 0.970
##
##
## [1] "C18.0"
```

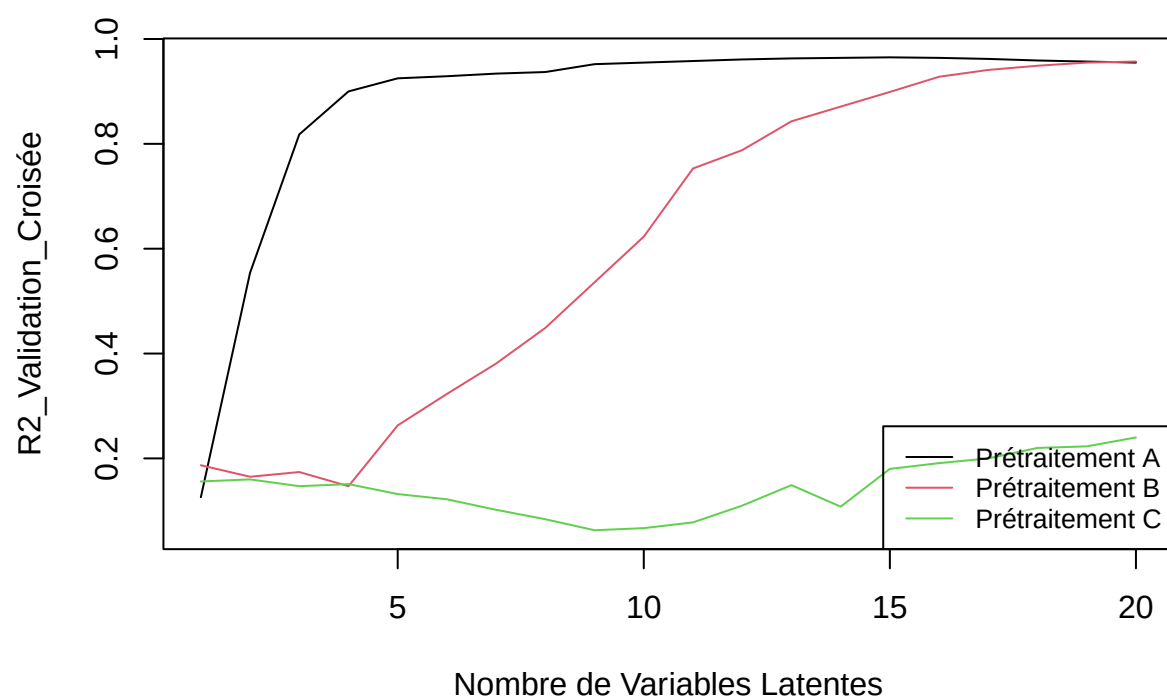


```
## [1] "Prétraitement A"
## [1] "R2"
## [1] 1.000 0.035 0.154 0.203 0.241 0.246 0.243 0.232 0.218 0.166 0.138 0.117
## [13] 0.116 0.130 0.214 0.198 0.200 0.216 0.220 0.190 0.210
## [1] "SEP"
## [1] 0.751 0.733 0.689 0.670 0.656 0.656 0.661 0.669 0.682 0.720 0.758 0.783
## [13] 0.791 0.785 0.733 0.742 0.756 0.752 0.760 0.801 0.793
##
##
## [1] "C18.1n9"
```

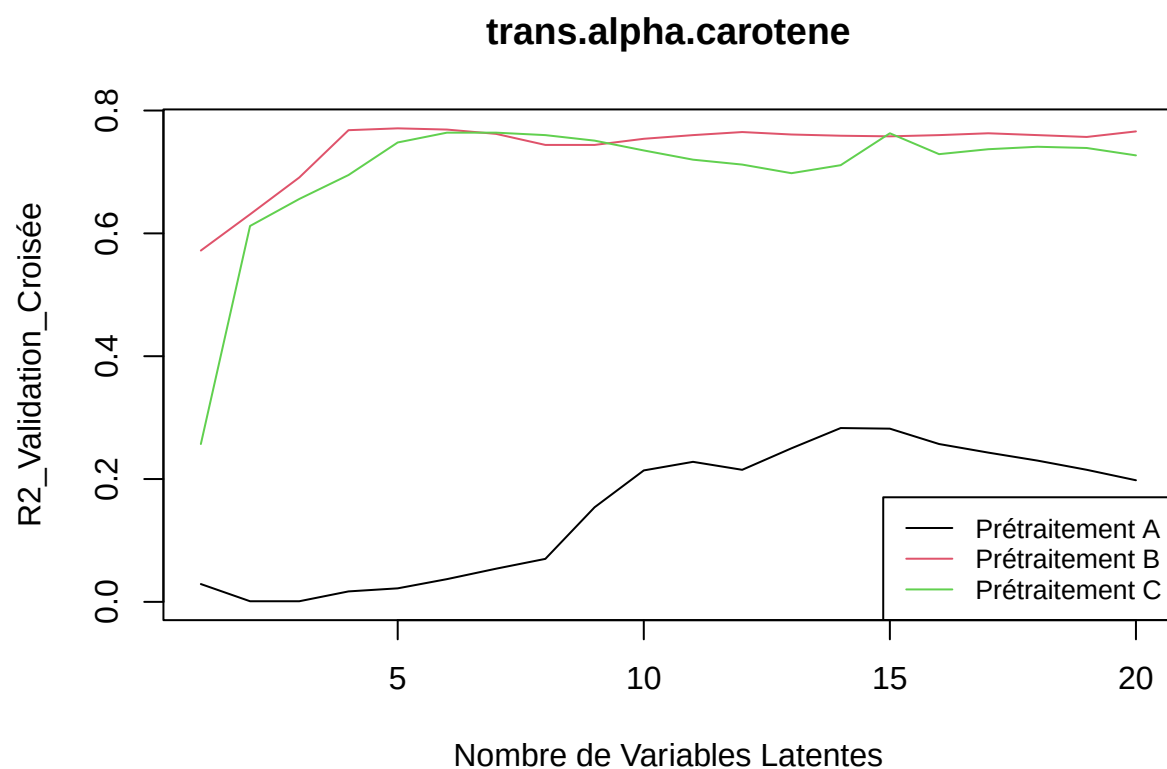


```
## [1] "Prétraitement A"
## [1] "R2"
## [1] 1.000 0.171 0.553 0.811 0.920 0.951 0.953 0.955 0.958 0.960 0.965 0.966
## [13] 0.969 0.971 0.971 0.971 0.972 0.970 0.968 0.968 0.969
## [1] "SEP"
## [1] 3.045 2.756 2.018 1.314 0.856 0.670 0.657 0.643 0.617 0.603 0.569 0.556
## [13] 0.532 0.516 0.512 0.514 0.509 0.528 0.539 0.539 0.536
##
##
## [1] "C18.2"
```

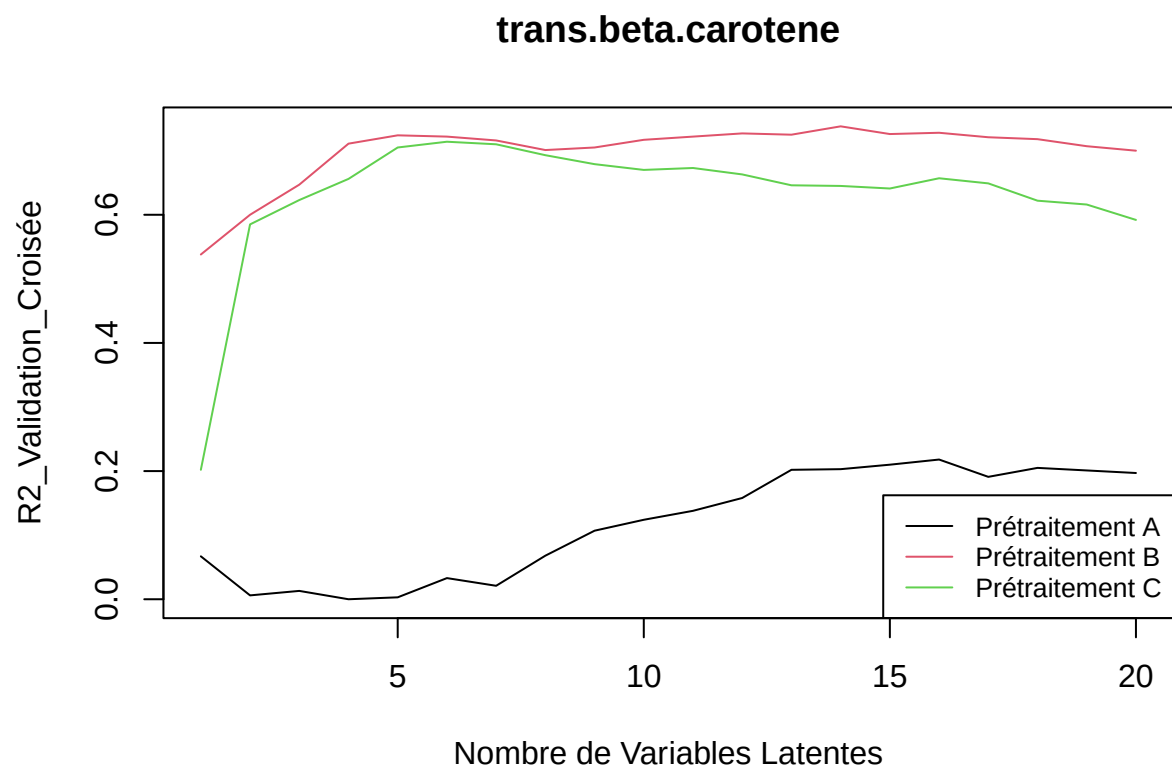
C18.2



```
## [1] "Prétraitement A"
## [1] "R2"
## [1] 1.000 0.126 0.554 0.818 0.900 0.925 0.929 0.934 0.937 0.952 0.955 0.958
## [13] 0.961 0.963 0.964 0.965 0.964 0.962 0.959 0.957 0.955
## [1] "SEP"
## [1] 1.152 1.078 0.763 0.490 0.360 0.312 0.305 0.293 0.288 0.251 0.242 0.234
## [13] 0.224 0.221 0.215 0.213 0.217 0.224 0.231 0.236 0.244
##
##
## [1] "trans.alpha.carotene"
```

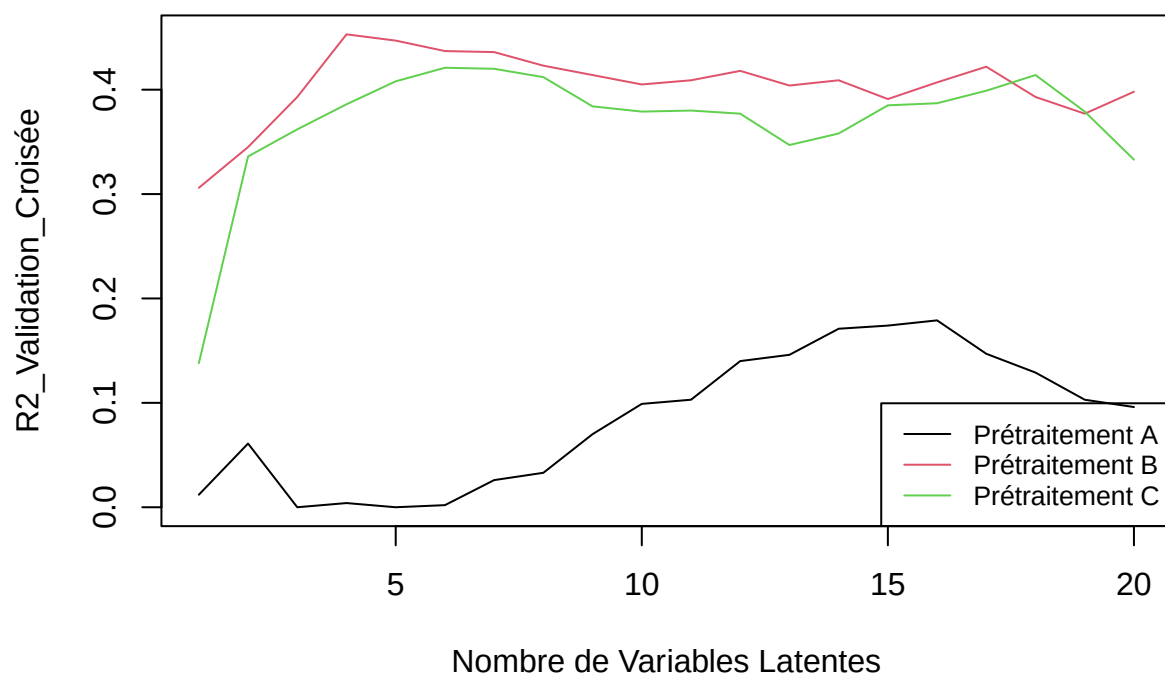


```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.572 0.631 0.691 0.768 0.771 0.769 0.762 0.744 0.744 0.754 0.760
## [13] 0.765 0.761 0.759 0.758 0.760 0.763 0.760 0.757 0.766
## [1] "SEP"
## [1] 173.388 112.473 104.407 95.595 82.818 82.425 82.965 84.369 88.265
## [10] 88.366 86.543 85.469 84.765 86.051 86.640 87.008 86.937 86.414
## [19] 87.231 88.272 86.111
##
##
## [1] "trans.beta.carotene"
```



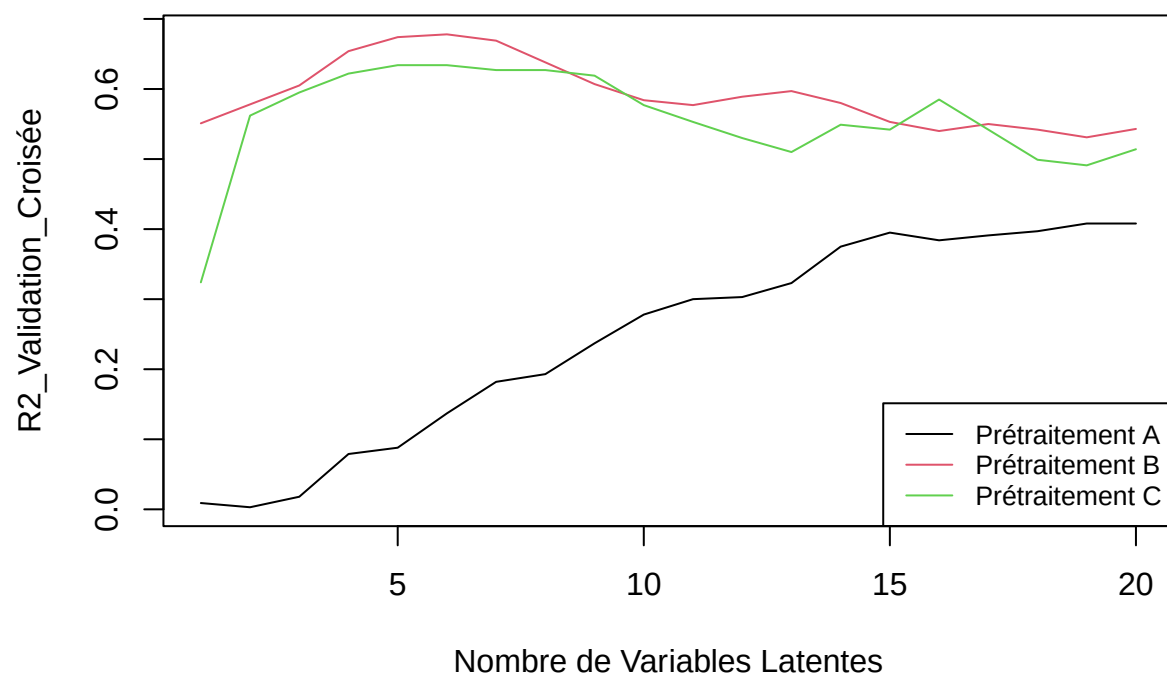
```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.538 0.600 0.647 0.711 0.724 0.722 0.716 0.701 0.705 0.717 0.722
## [13] 0.727 0.725 0.738 0.726 0.728 0.721 0.718 0.707 0.700
## [1] "SEP"
## [1] 238.262 160.629 149.405 140.464 127.119 124.529 125.454 127.221 131.610
## [10] 130.951 127.872 126.834 125.916 127.157 124.093 127.509 127.083 129.713
## [19] 130.745 133.867 136.338
##
##
## [1] "X13.cis.beta.carotene"
```


X13.cis.beta.carotene

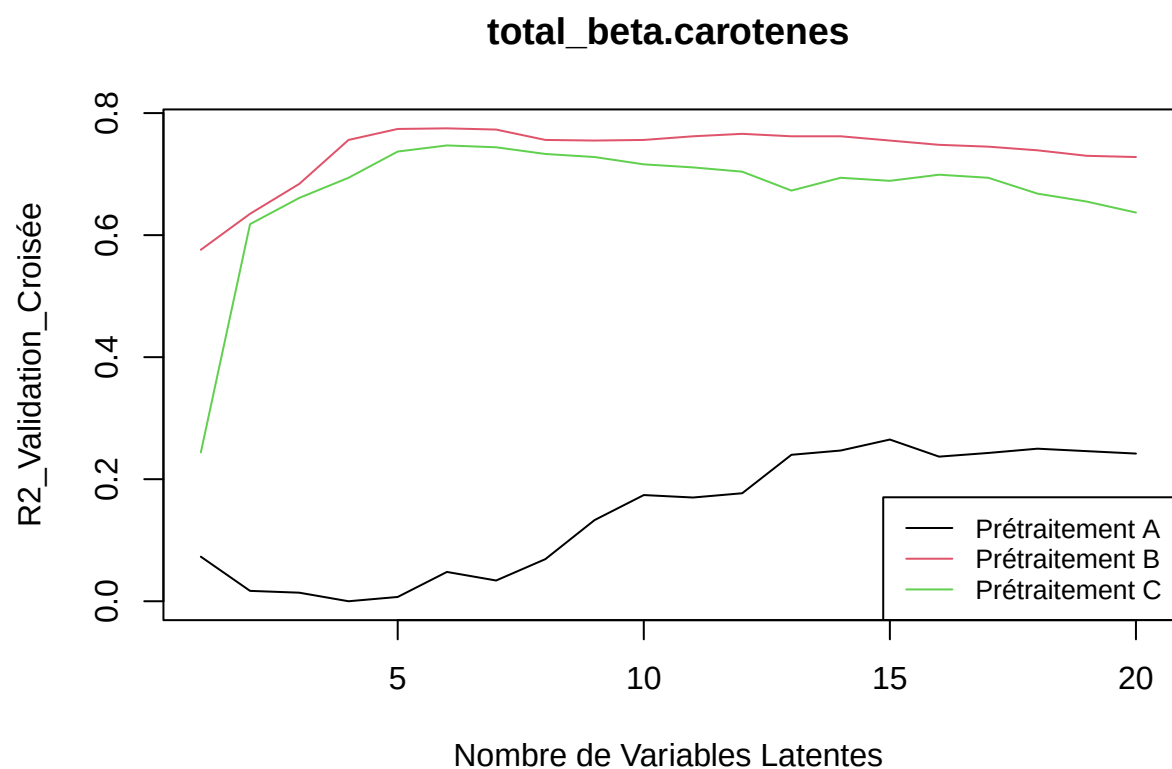


```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.306 0.345 0.393 0.453 0.447 0.437 0.436 0.423 0.414 0.405 0.409
## [13] 0.418 0.404 0.409 0.391 0.407 0.422 0.393 0.377 0.398
## [1] "SEP"
## [1] 38.179 31.532 30.654 29.540 28.154 28.533 29.077 29.231 29.889 30.330
## [11] 30.843 30.812 30.512 31.373 31.451 32.497 31.931 31.356 32.686 33.592
## [21] 32.941
##
##
## [1] "X9.cis.beta.carotene"
```

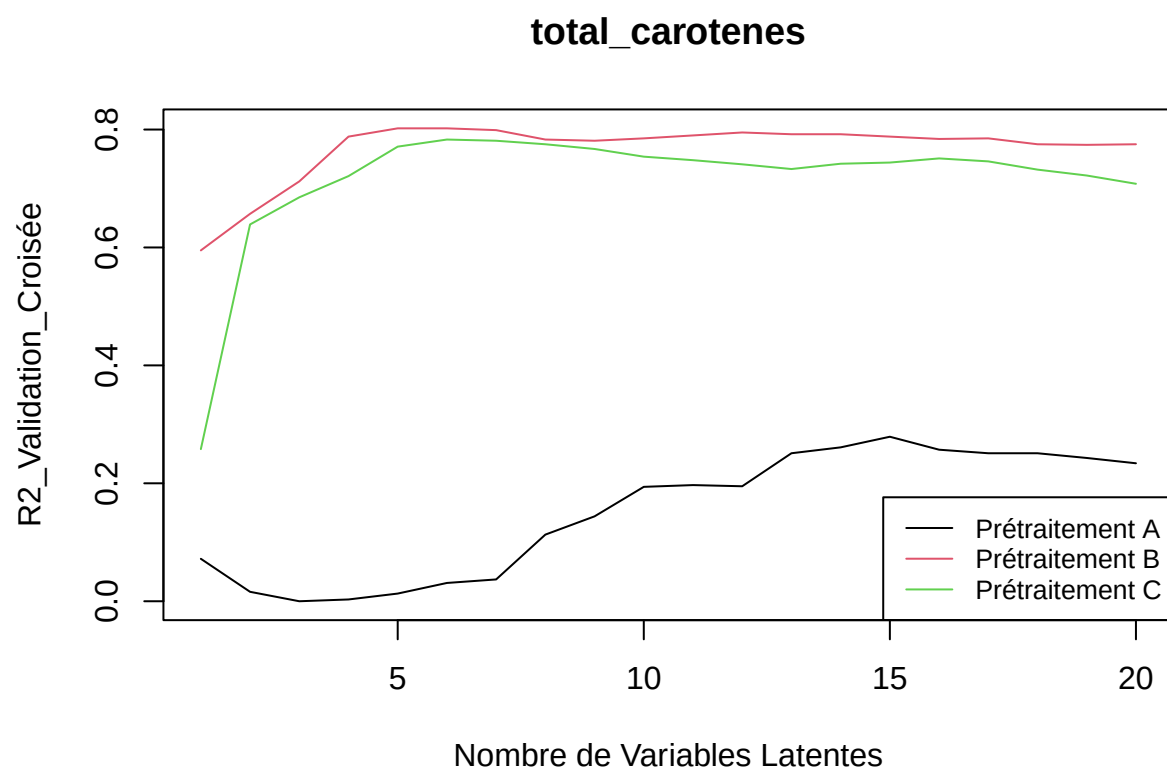
X9.cis.beta.carotene



```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.551 0.578 0.605 0.654 0.674 0.678 0.669 0.638 0.607 0.584 0.577
## [13] 0.589 0.597 0.580 0.553 0.540 0.550 0.542 0.531 0.543
## [1] "SEP"
## [1] 64.424 42.794 41.471 40.157 37.619 36.565 36.435 36.983 38.998 41.132
## [11] 42.718 43.249 42.557 42.183 43.442 45.283 46.329 45.702 46.293 47.155
## [21] 46.481
##
##
## [1] "total_beta.carotenes"
```

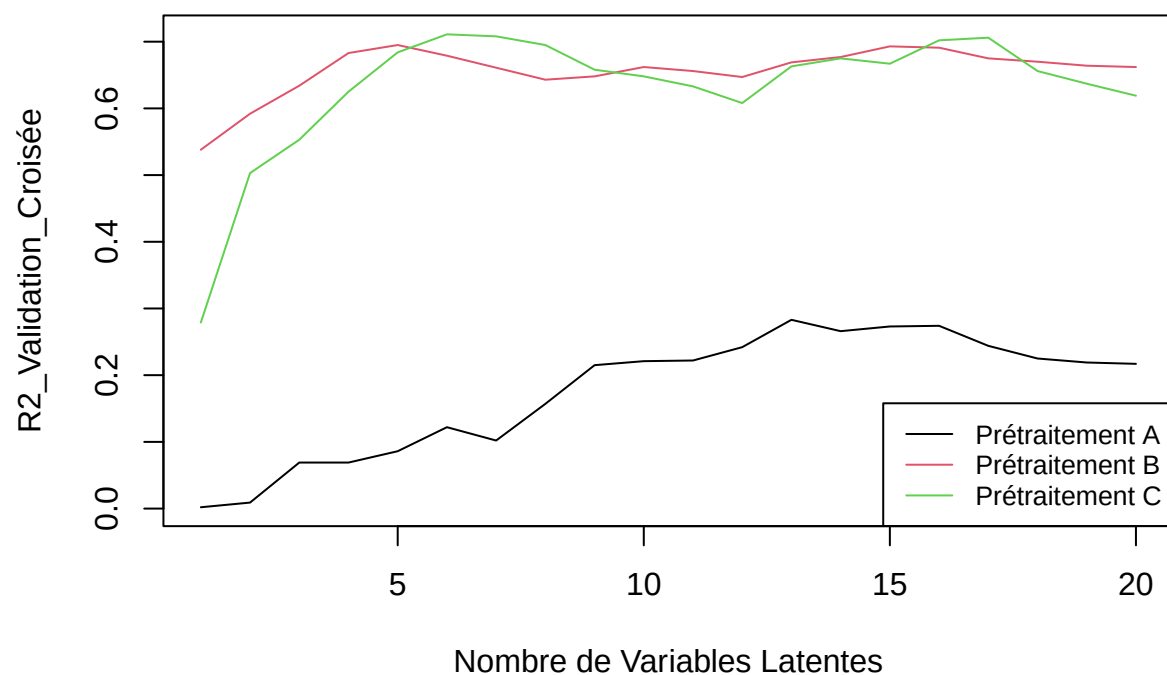


```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.576 0.635 0.684 0.756 0.774 0.775 0.773 0.756 0.755 0.756 0.762
## [13] 0.766 0.762 0.762 0.755 0.748 0.745 0.739 0.730 0.728
## [1] "SEP"
## [1] 321.436 207.594 192.584 179.014 157.606 151.729 152.047 153.027 159.760
## [10] 160.592 160.284 158.407 157.142 159.306 159.725 162.484 165.358 167.319
## [19] 170.279 173.636 174.821
##
##
## [1] "total_carotenes"
```

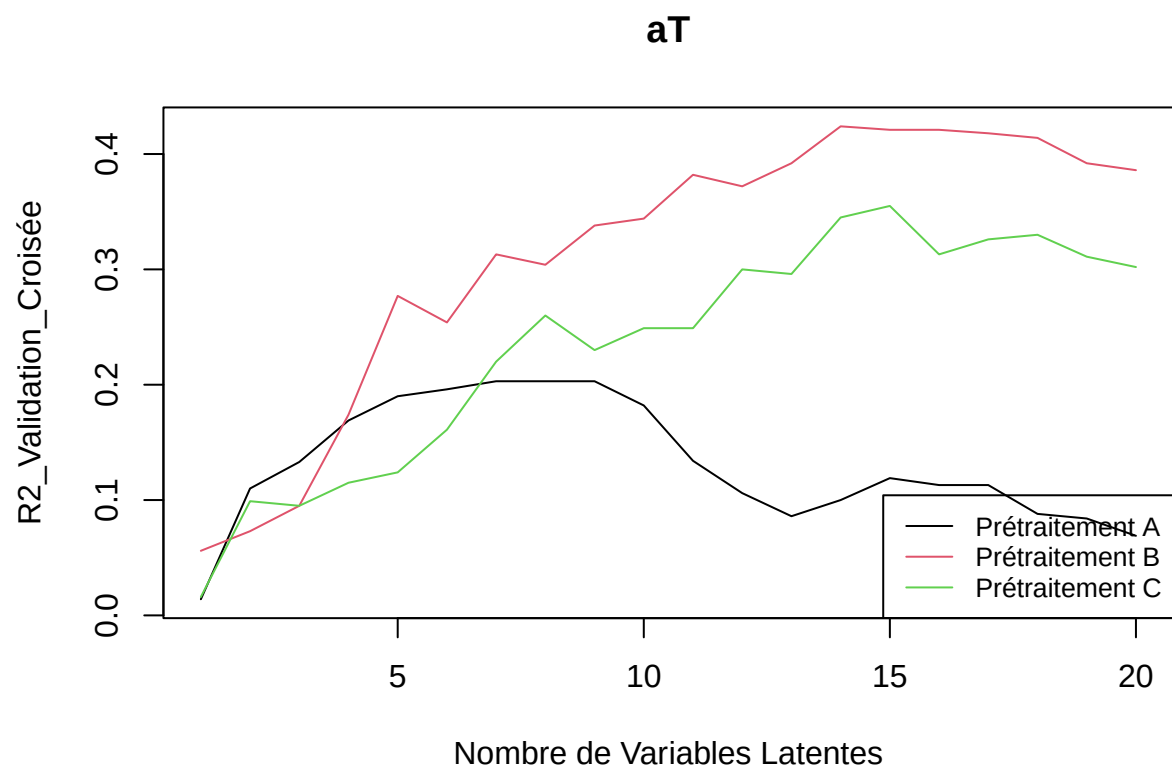


```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.595 0.657 0.712 0.788 0.802 0.802 0.799 0.783 0.781 0.785 0.790
## [13] 0.795 0.792 0.792 0.788 0.784 0.785 0.775 0.774 0.775
## [1] "SEP"
## [1] 486.520 306.734 282.540 258.831 222.471 215.193 215.862 217.620 227.981
## [10] 229.395 227.247 224.640 221.925 224.493 225.256 227.889 230.854 231.478
## [19] 237.464 238.567 237.784
##
##
## [1] "Lycopene"
```

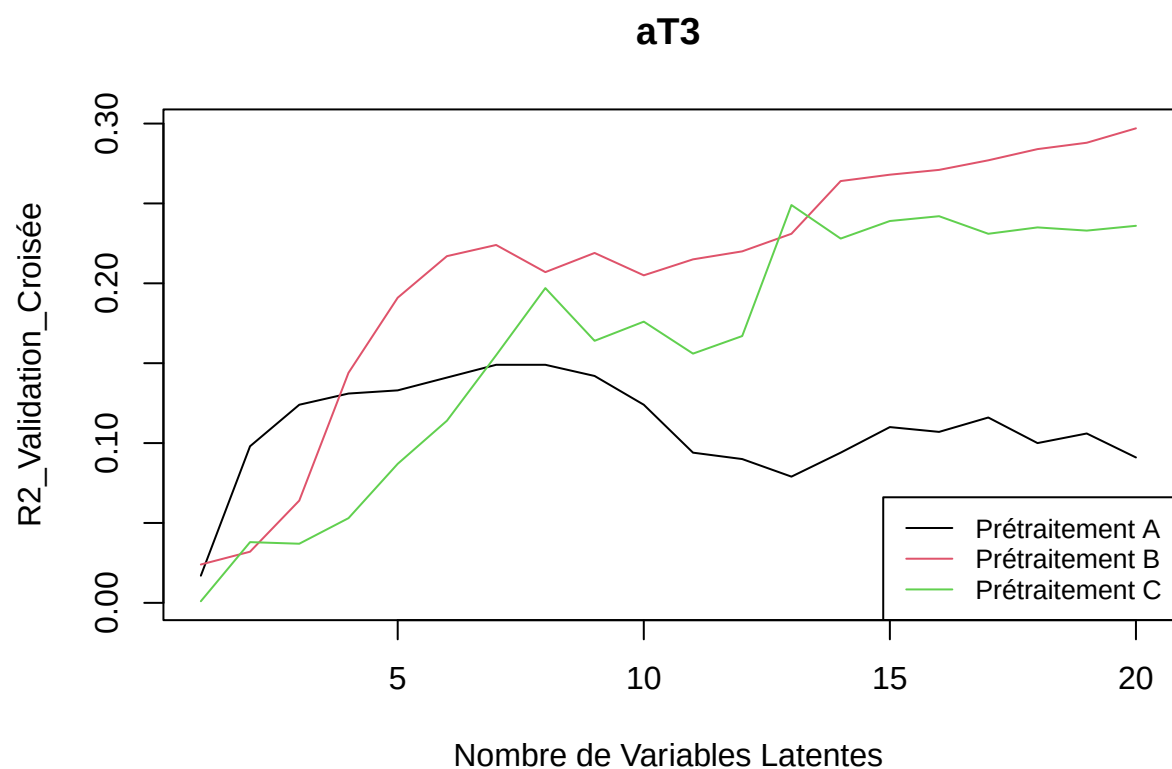
Lycopene



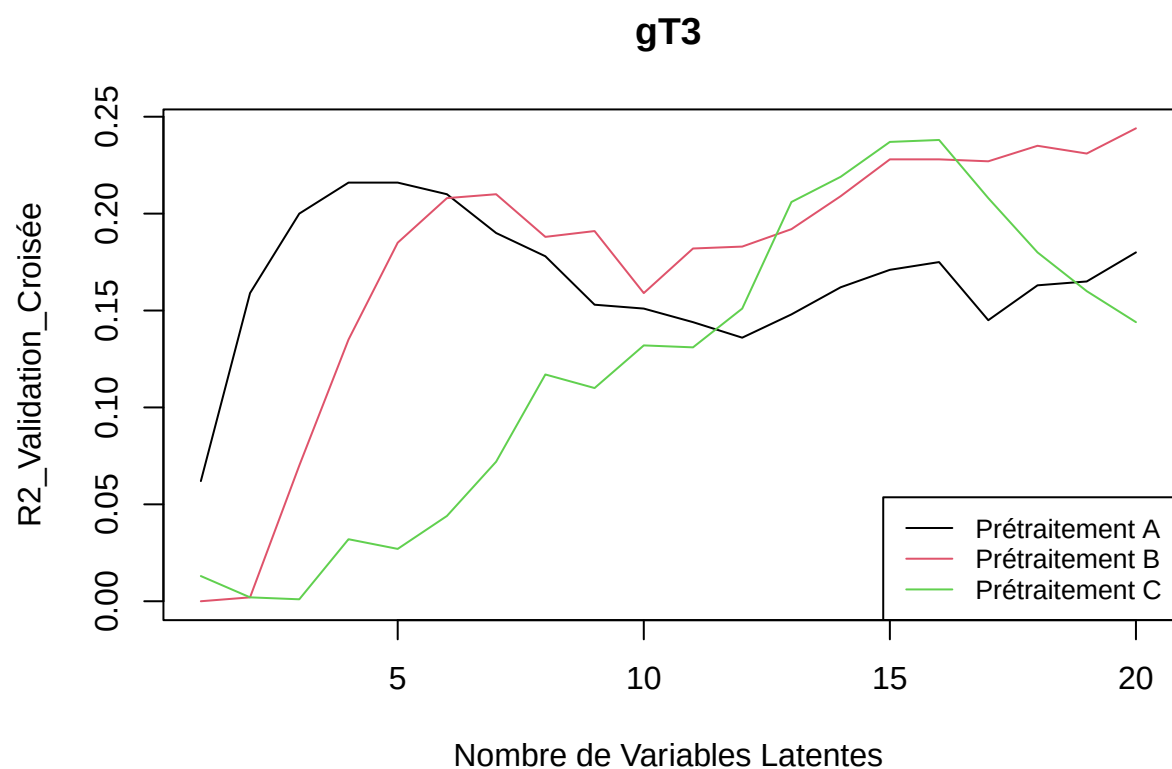
```
## [1] "Prétraitement C"
## [1] "R2"
## [1] 1.000 0.279 0.503 0.553 0.625 0.684 0.711 0.708 0.695 0.658 0.648 0.633
## [13] 0.608 0.663 0.675 0.667 0.702 0.706 0.656 0.637 0.619
## [1] "SEP"
## [1] 6.487 5.465 4.535 4.302 3.941 3.616 3.464 3.487 3.571 3.815 3.893 3.998
## [13] 4.158 3.793 3.732 3.802 3.593 3.565 3.901 4.027 4.184
##
##
## [1] "aT"
```



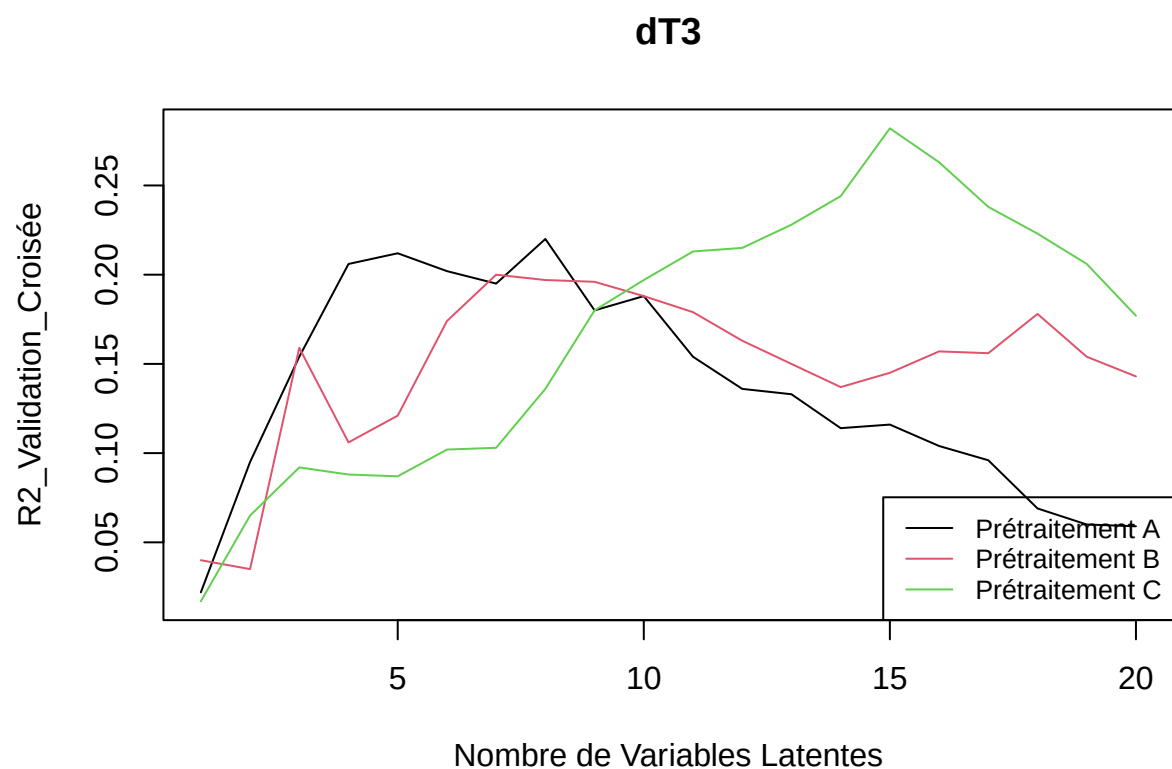
```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.056 0.073 0.095 0.174 0.277 0.254 0.313 0.304 0.338 0.344 0.382
## [13] 0.372 0.392 0.424 0.421 0.421 0.418 0.414 0.392 0.386
## [1] "SEP"
## [1] 95.576 92.396 91.603 90.757 87.520 81.926 83.807 80.496 81.455 80.634
## [11] 79.803 77.858 78.217 77.219 75.281 75.278 76.346 76.377 78.072 79.969
## [21] 80.743
##
##
## [1] "aT3"
```



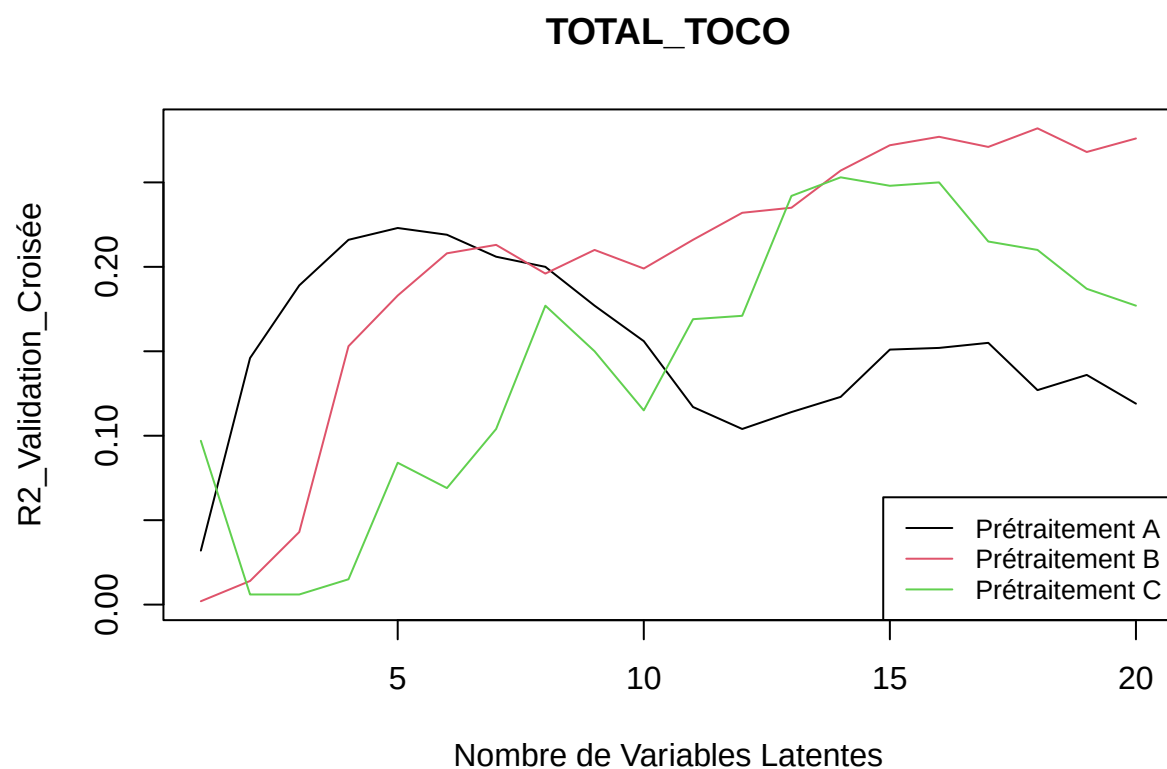
```
## [1] "Prétraitement C"
## [1] "R2"
## [1] 1.000 0.001 0.038 0.037 0.053 0.087 0.114 0.155 0.197 0.164 0.176 0.156
## [13] 0.167 0.249 0.228 0.239 0.242 0.231 0.235 0.233 0.236
## [1] "SEP"
## [1] 107.204 106.938 104.941 105.444 105.359 103.445 101.789 99.685 96.463
## [10] 99.740 99.547 103.081 102.143 95.536 98.942 98.309 98.656 100.273
## [19] 103.716 102.988 104.350
##
##
## [1] "gT3"
```



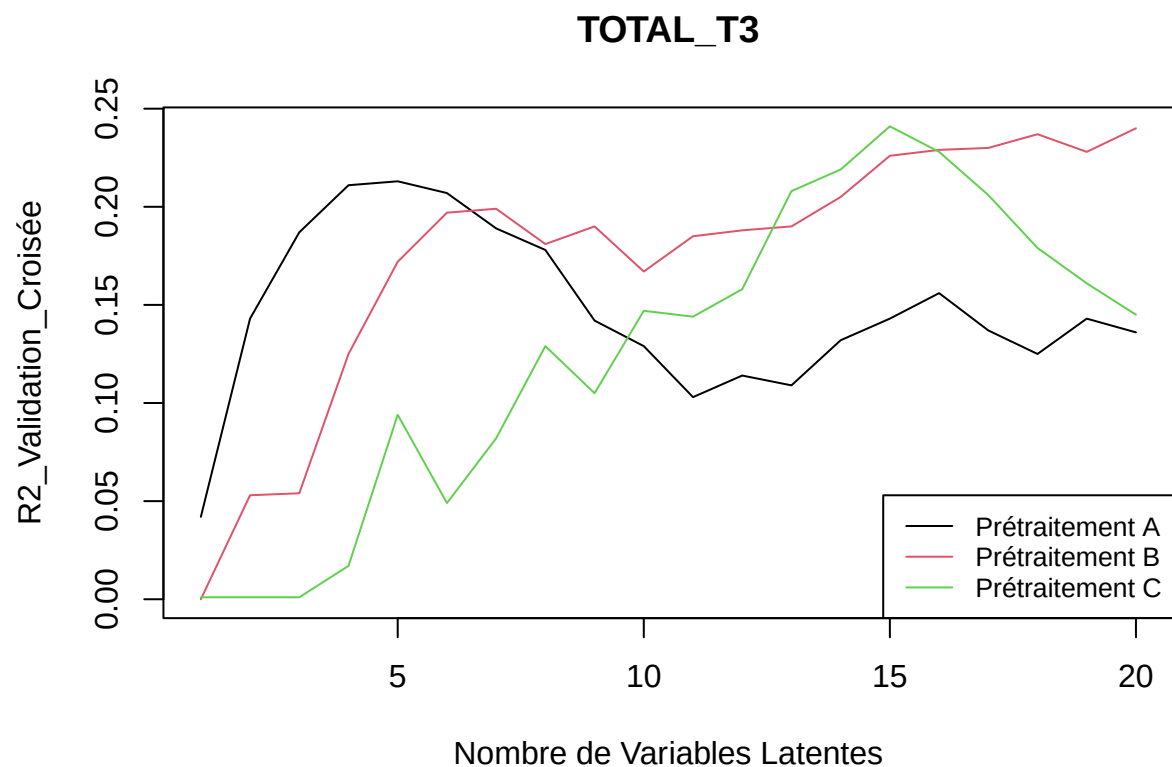
```
## [1] "Prétraitement A"
## [1] "R2"
## [1] 1.000 0.062 0.159 0.200 0.216 0.216 0.210 0.190 0.178 0.153 0.151 0.144
## [13] 0.136 0.148 0.162 0.171 0.175 0.145 0.163 0.165 0.180
## [1] "SEP"
## [1] 120.143 116.026 109.790 107.318 107.069 107.342 108.181 111.311 113.148
## [10] 117.610 119.763 123.562 123.794 124.113 123.167 123.932 123.658 130.140
## [19] 130.288 131.280 130.340
##
##
## [1] "dT3"
```

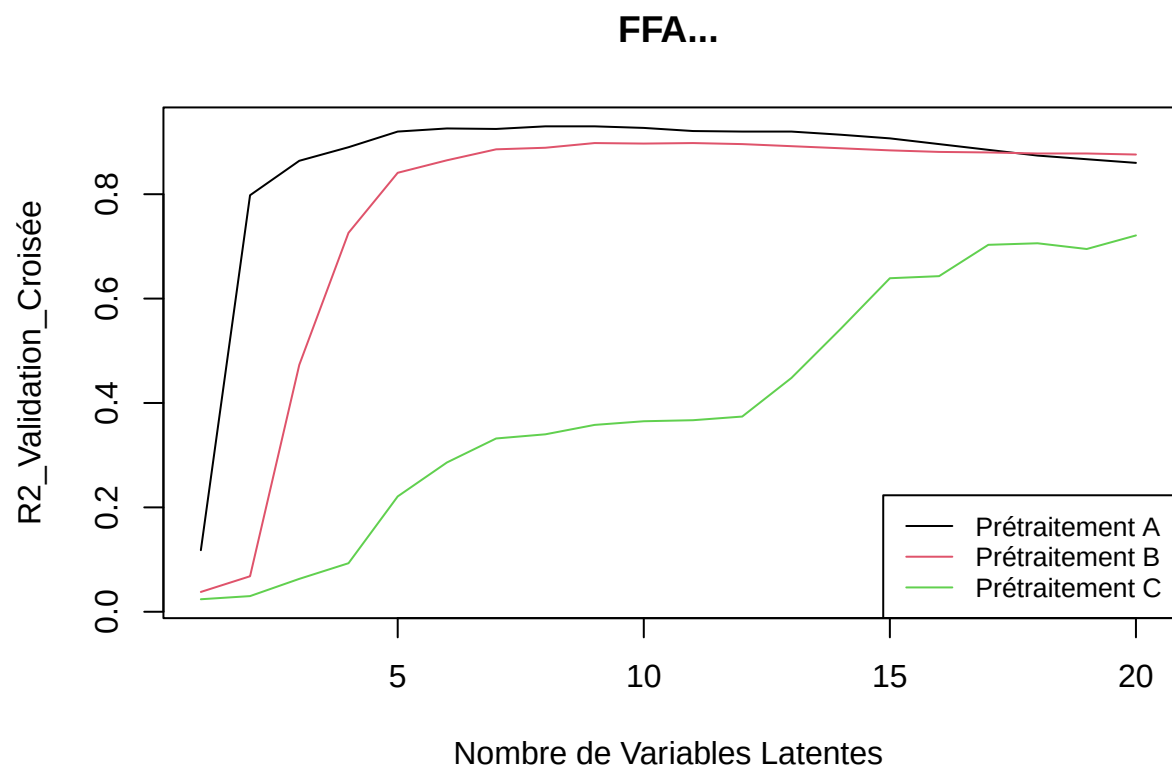
```
## [1] "Prétraitement C"
## [1] "R2"
## [1] 1.000 0.017 0.065 0.092 0.088 0.087 0.102 0.103 0.136 0.180 0.197 0.213
## [13] 0.215 0.228 0.244 0.282 0.263 0.238 0.223 0.206 0.177
## [1] "SEP"
## [1] 54.303 53.527 52.325 51.848 51.967 52.134 51.812 52.384 51.212 49.900
## [11] 49.525 48.859 49.036 49.338 49.437 47.797 49.385 51.555 53.309 54.594
## [21] 58.011
##
##
## [1] "TOTAL_TOCO"
```



```
## [1] "Prétraitement C"
## [1] "R2"
## [1] 1.000 0.097 0.006 0.006 0.015 0.084 0.069 0.104 0.177 0.150 0.115 0.169
## [13] 0.171 0.242 0.253 0.248 0.250 0.215 0.210 0.187 0.177
## [1] "SEP"
## [1] 321.455 342.761 321.462 326.061 327.610 308.824 314.187 310.550 292.363
## [10] 302.404 317.882 302.877 305.497 288.200 288.107 291.345 293.771 307.454
## [19] 320.011 329.280 339.939
##
##
## [1] "TOTAL_T3"
```

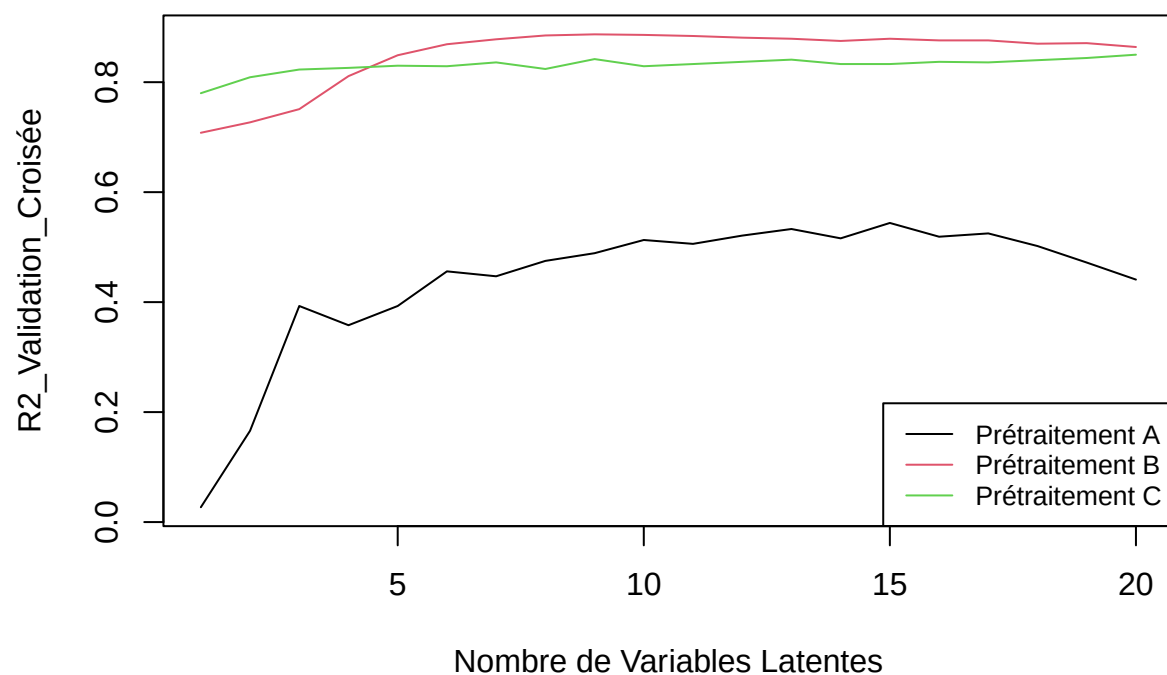


```
## [1] "Prétraitement A"
## [1] "R2"
## [1] 1.000 0.042 0.143 0.187 0.211 0.213 0.207 0.189 0.178 0.142 0.129 0.103
## [13] 0.114 0.109 0.132 0.143 0.156 0.137 0.125 0.143 0.136
## [1] "SEP"
## [1] 243.445 238.045 224.695 219.440 217.755 218.129 219.852 225.499 228.738
## [10] 240.666 248.299 265.427 261.582 264.899 258.146 258.562 255.850 265.962
## [19] 275.502 273.886 279.408
##
##
## [1] "FFA..."
```



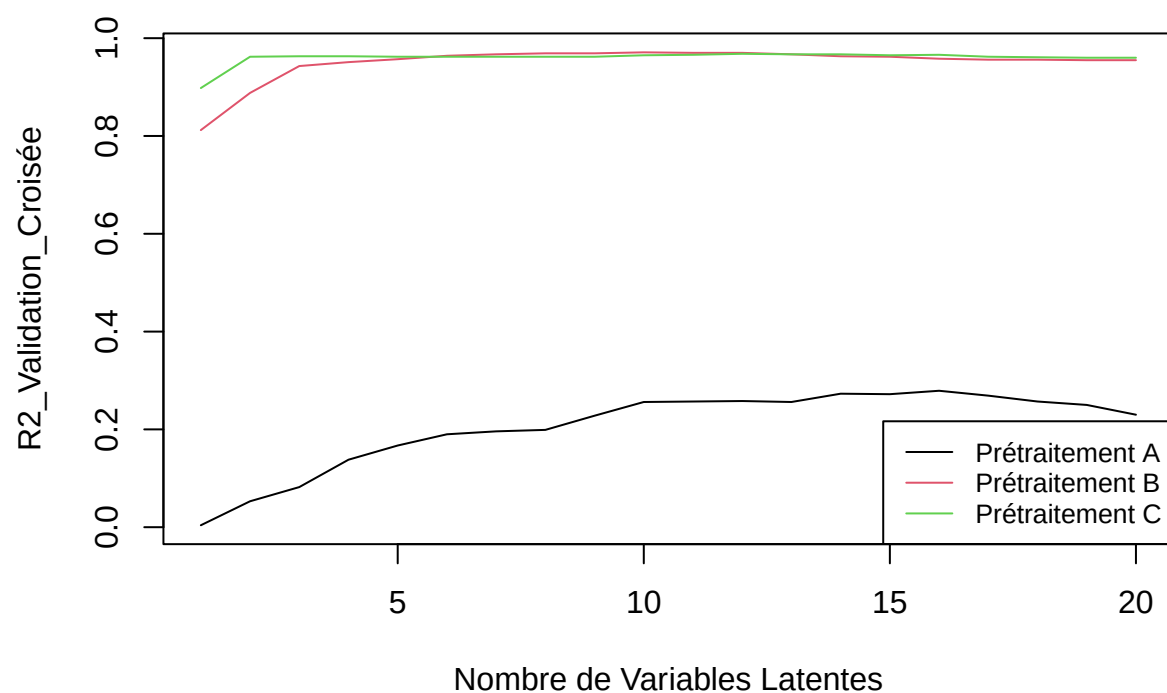
```
## [1] "Prétraitement A"
## [1] "R2"
## [1] 1.000 0.118 0.798 0.864 0.890 0.920 0.926 0.925 0.930 0.930 0.927 0.921
## [13] 0.920 0.920 0.914 0.907 0.896 0.885 0.874 0.867 0.860
## [1] "SEP"
## [1] 6.162 5.748 2.752 2.260 2.025 1.722 1.666 1.669 1.610 1.621 1.648 1.719
## [13] 1.727 1.732 1.792 1.868 1.980 2.080 2.188 2.254 2.326
##
##
## [1] "L."
```

L.



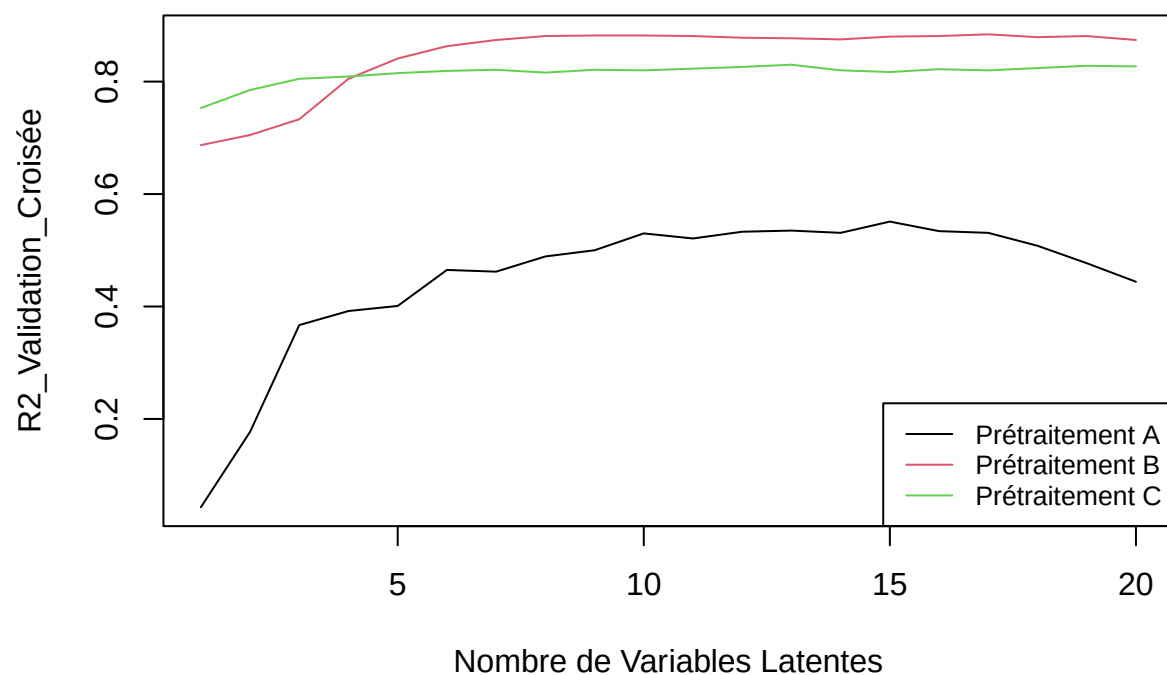
```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.708 0.727 0.751 0.811 0.849 0.869 0.878 0.885 0.887 0.886 0.884
## [13] 0.881 0.879 0.875 0.879 0.876 0.876 0.870 0.871 0.864
## [1] "SEP"
## [1] 2.826 1.514 1.465 1.397 1.217 1.089 1.016 0.980 0.952 0.946 0.950 0.958
## [13] 0.976 0.986 1.006 0.989 1.001 1.003 1.030 1.029 1.061
##
##
## [1] "a."
```

a.



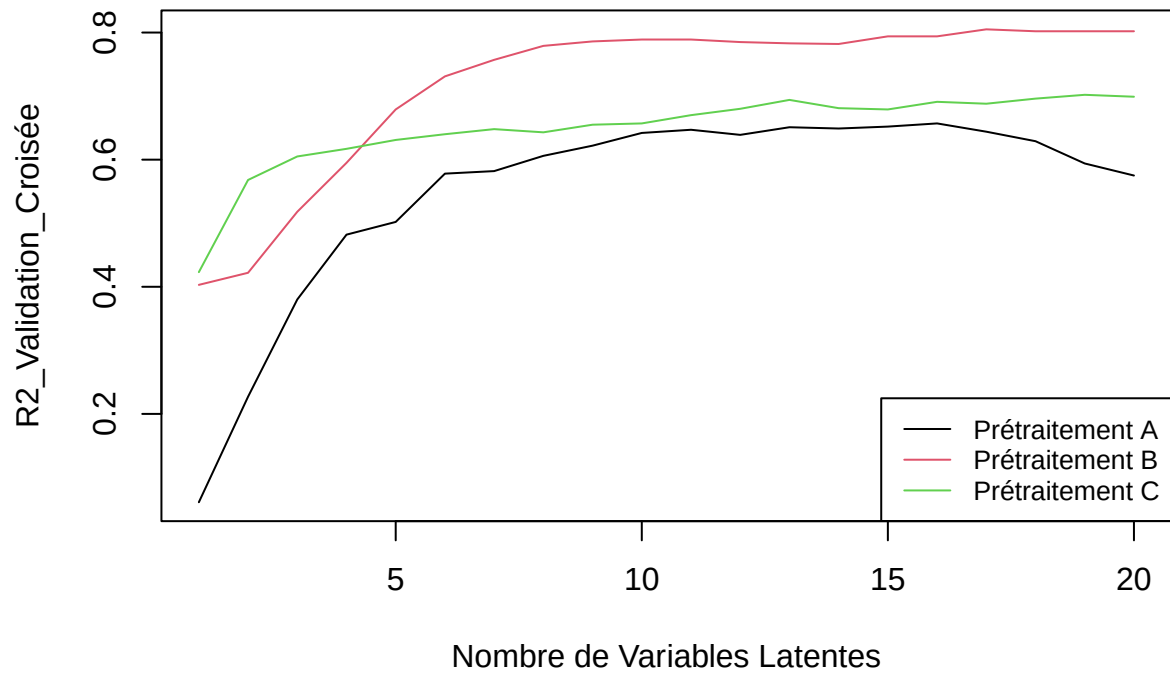
```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.812 0.888 0.943 0.951 0.957 0.964 0.967 0.969 0.969 0.971 0.970
## [13] 0.970 0.967 0.963 0.962 0.958 0.956 0.956 0.955 0.955
## [1] "SEP"
## [1] 4.003 1.720 1.331 0.949 0.876 0.824 0.755 0.724 0.702 0.694 0.674 0.691
## [13] 0.693 0.721 0.761 0.775 0.816 0.838 0.839 0.844 0.843
##
##
## [1] "b."
```

b.

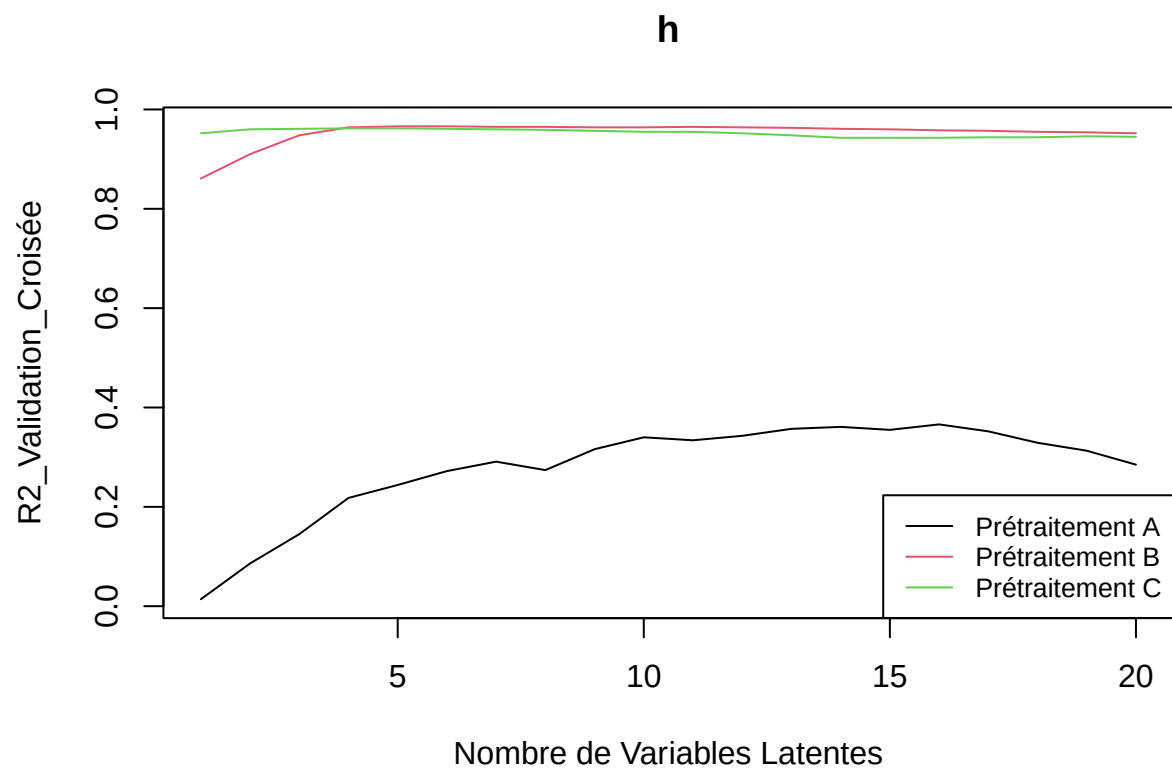


```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.687 0.705 0.733 0.805 0.841 0.863 0.874 0.881 0.882 0.882 0.881
## [13] 0.878 0.877 0.875 0.880 0.881 0.884 0.879 0.881 0.874
## [1] "SEP"
## [1] 4.900 2.718 2.639 2.511 2.145 1.937 1.802 1.729 1.680 1.670 1.675 1.685
## [13] 1.711 1.721 1.741 1.702 1.700 1.682 1.719 1.704 1.765
##
##
## [1] "C."
```

C.



```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.403 0.422 0.518 0.595 0.679 0.731 0.757 0.779 0.786 0.789 0.789
## [13] 0.785 0.783 0.782 0.794 0.794 0.805 0.802 0.802 0.802
## [1] "SEP"
## [1] 3.264 2.501 2.461 2.260 2.063 1.836 1.682 1.599 1.525 1.504 1.498 1.500
## [13] 1.522 1.532 1.545 1.500 1.499 1.462 1.474 1.481 1.488
##
##
## [1] "h"
```

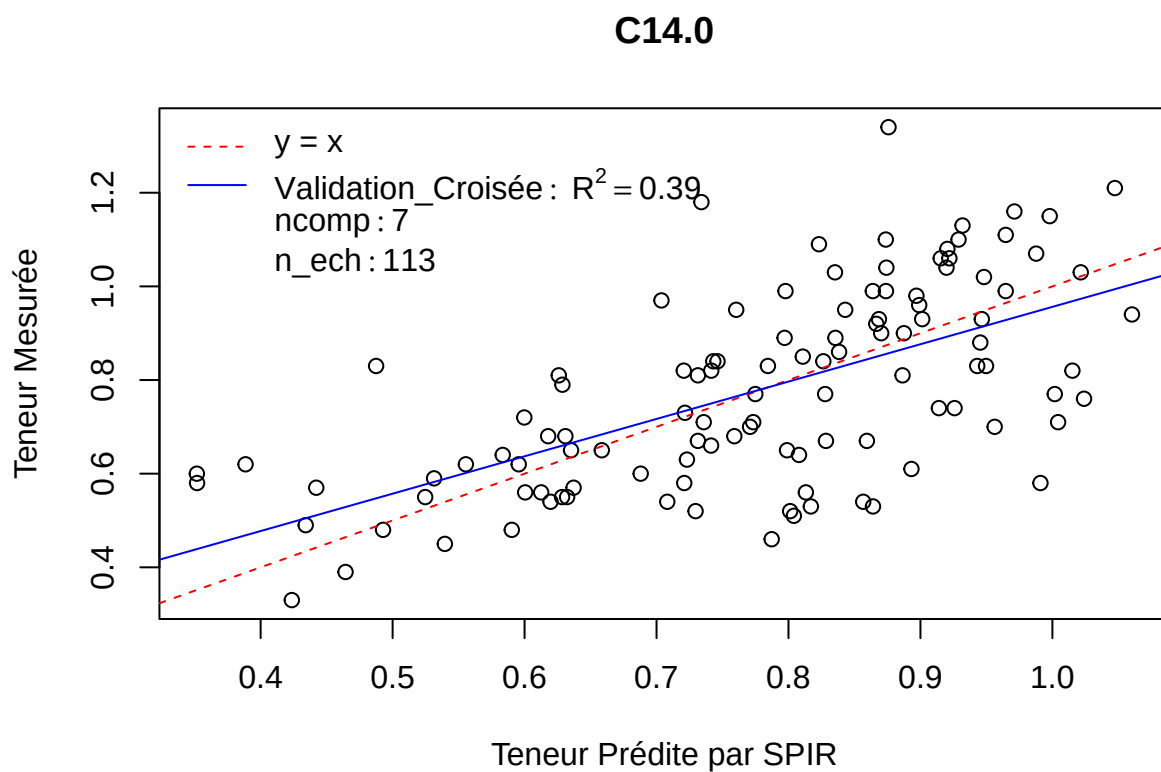
```
## [1] "Prétraitement B"
## [1] "R2"
## [1] 1.000 0.861 0.910 0.948 0.964 0.966 0.966 0.965 0.965 0.964 0.964 0.965
## [13] 0.964 0.963 0.961 0.960 0.958 0.957 0.955 0.954 0.952
## [1] "SEP"
## [1] 8.003 2.955 2.381 1.809 1.502 1.471 1.463 1.475 1.494 1.497 1.497 1.493
## [13] 1.501 1.530 1.572 1.597 1.624 1.640 1.682 1.712 1.740
```

Résultats ASD DIADE (défini manuellement après observation des courbes de CV)

Propriete	pre	nVL
C14.0	adj , red , snv , sder, , 950 , 750 , 1 , , 2 , 3 , 15	7
C16.0	adj , red , snv , sder, , 950 , 750 , 1 , , 2 , 3 , 15	5
C18.0	adj , red , snv , sder, , 950 , 750 , 1 , , 2 , 3 , 15	5
C18.1n9	adj , red , snv , sder, , 950 , 750 , 1 , , 2 , 3 , 15	5
C18.2	adj , red , snv , sder, , 950 , 750 , 1 , , 2 , 3 , 15	9
trans.alpha.carotene	adj , snv , sder, , , 2 , 3 , 15	4
trans.beta.carotene	adj , snv , sder, , , 2 , 3 , 15	5
X13.cis.beta.carotene	adj , snv , sder, , , 2 , 3 , 15	4
X9.cis.beta.carotene	adj , snv , sder, , , 2 , 3 , 15	6
total_beta.carotenes	adj , snv , sder, , , 2 , 3 , 15	5
total_carotenes	adj , snv , sder, , , 2 , 3 , 15	5
Lycopene	adj , snv , sder, , , 2 , 3 , 15	5
aT	adj , snv , sder, , , 2 , 3 , 15	14
aT3	adj , snv , sder, , , 2 , 3 , 15	14
gT3	adj , snv , sder, , , 2 , 3 , 15	15
dT3	adj , snv , sder, , , 2 , 3 , 15	7
TOTAL_TOCO	adj , snv , sder, , , 2 , 3 , 15	16
TOTAL_T3	adj , snv , sder, , , 2 , 3 , 15	18
FFA...	adj , red , snv , sder, , 950 , 750 , 1 , , 2 , 3 , 15	5
L.	adj , snv , sder, , , 2 , 3 , 15	8
a.	adj , red , snv , , 1 , 1300, 1 ,	3
b.	adj , snv , sder, , , 2 , 3 , 15	7
C.	adj , snv , sder, , , 2 , 3 , 15	10
h	adj , red , snv , , 1 , 1300, 1 ,	3

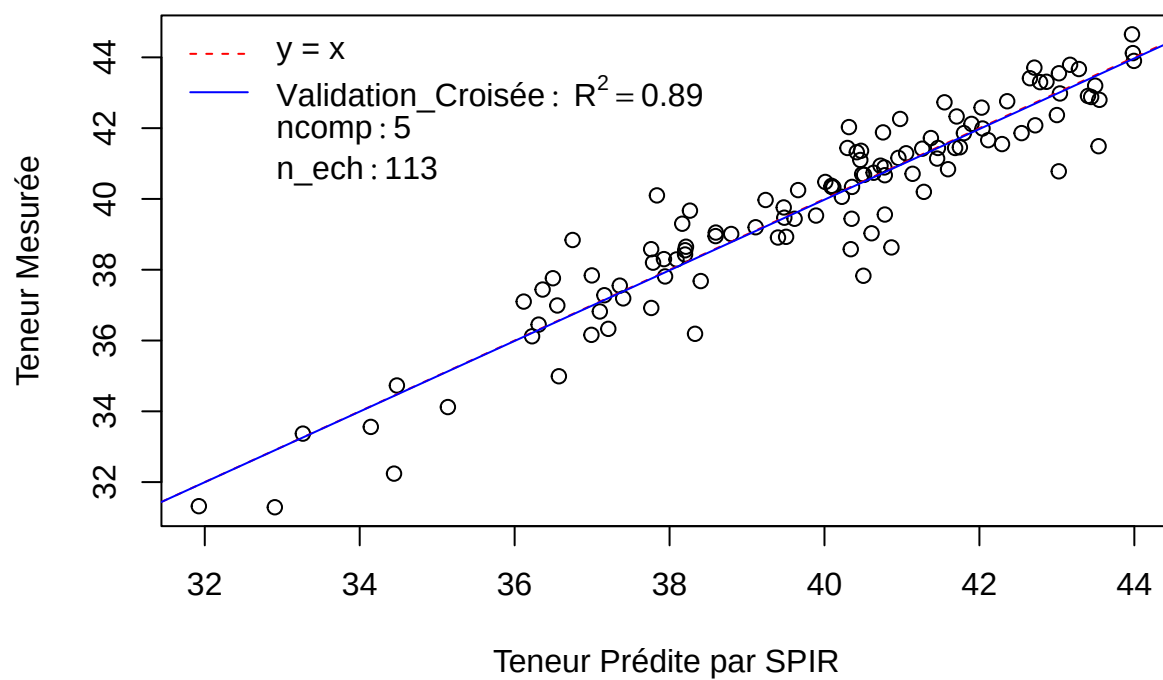
Resultat des Cross-Validation avec les bons paramètres (PRE+nVL)

```
## [1] "C14.0"  
## [1] "RANGE:1299-1750nm + SNV + DERIV.2nde (poly=3;window=3)"  
## [1] "R2 = 0.391"  
## [1] "SEP = 0.167"
```



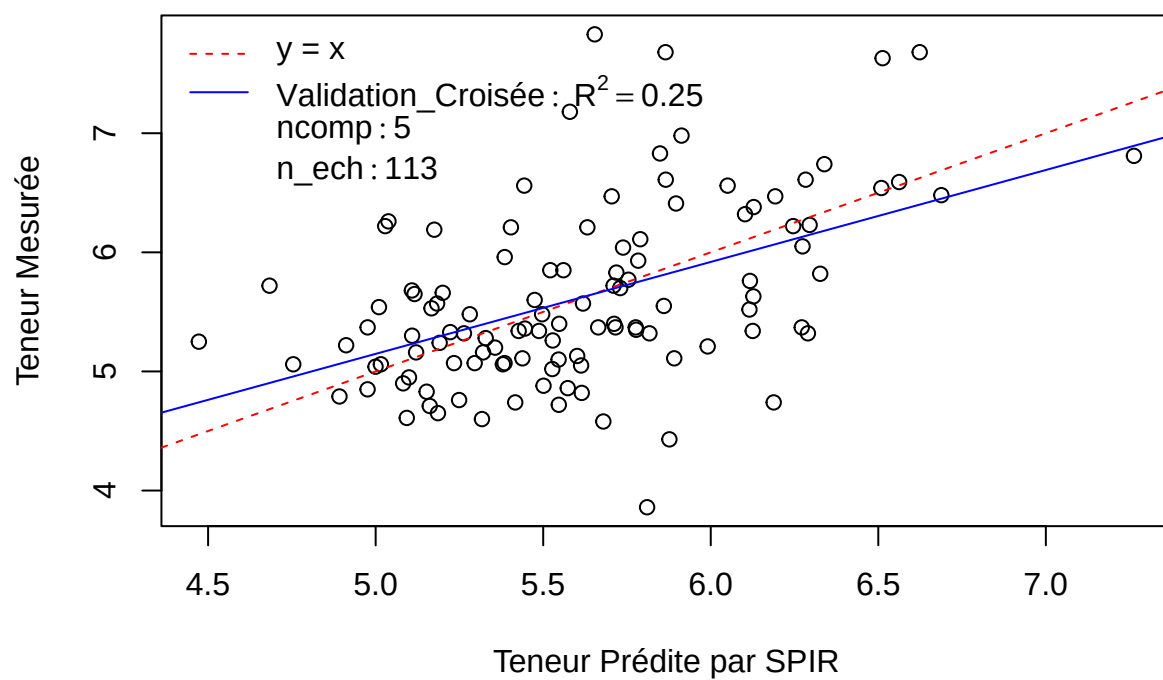
```
## [1] "C16.0"  
## [1] "RANGE:1299-1750nm + SNV + DERIV.2nde (poly=3;window=3)"  
## [1] "R2 = 0.895"  
## [1] "SEP = 0.894"
```

C16.0



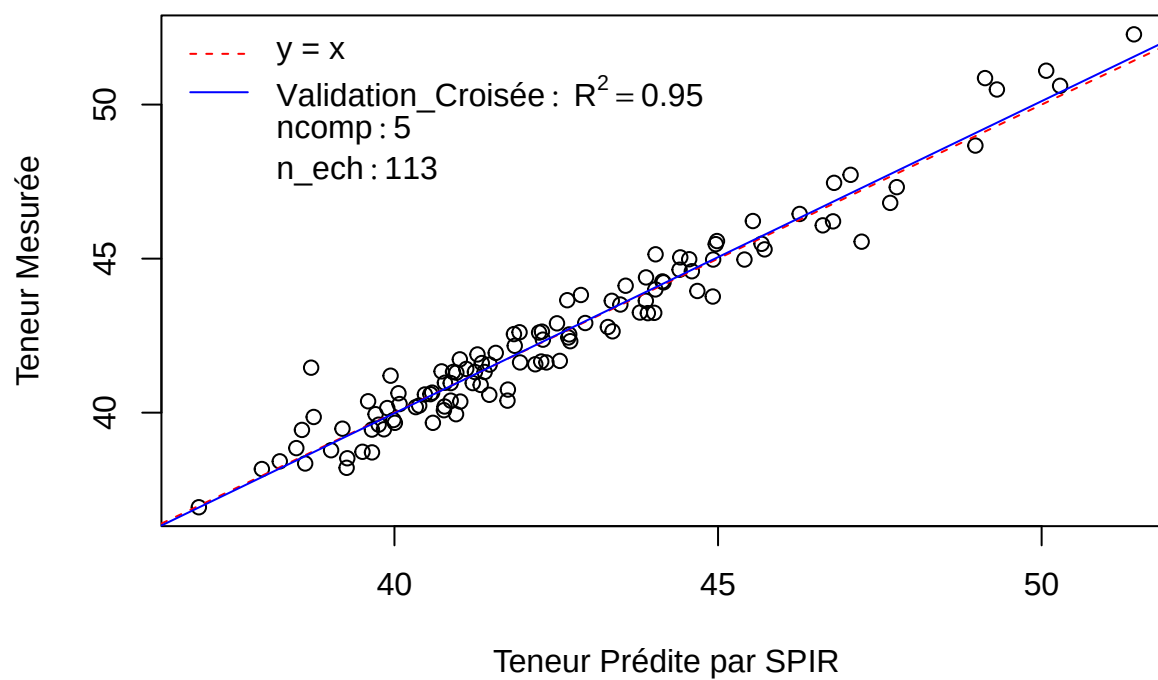
```
## [1] "C18.0"
## [1] "RANGE:1299-1750nm + SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.246"
## [1] "SEP = 0.656"
```

C18.0



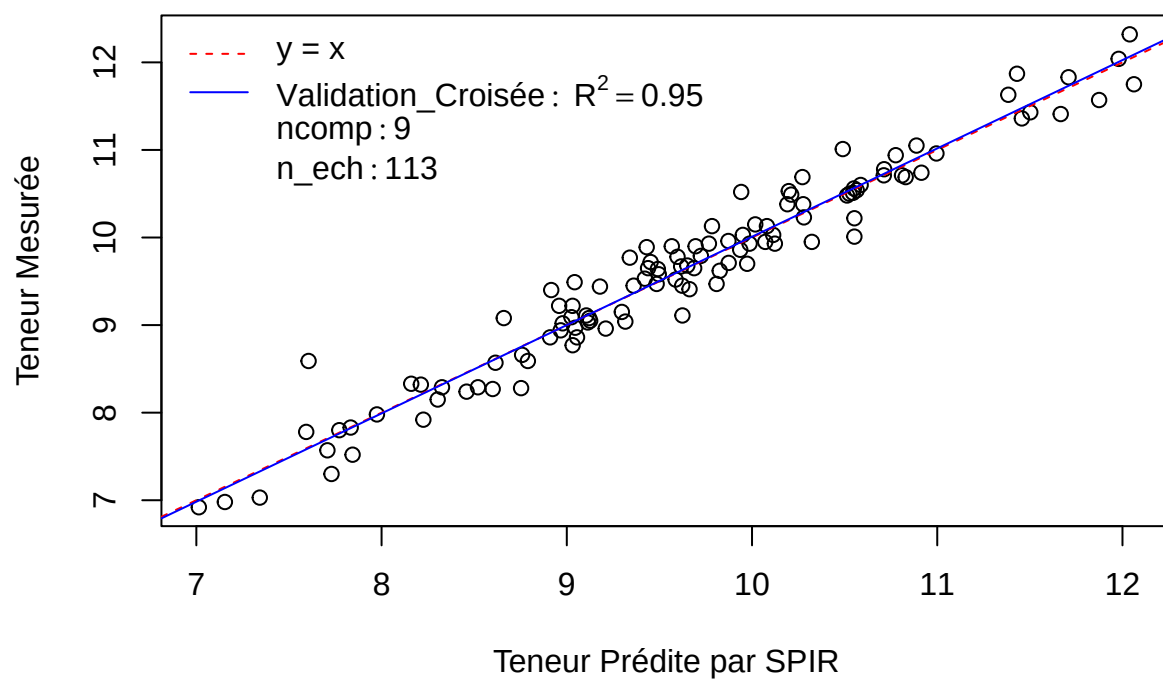
```
## [1] "C18.1n9"
## [1] "RANGE:1299-1750nm + SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.951"
## [1] "SEP = 0.67"
```

C18.1n9



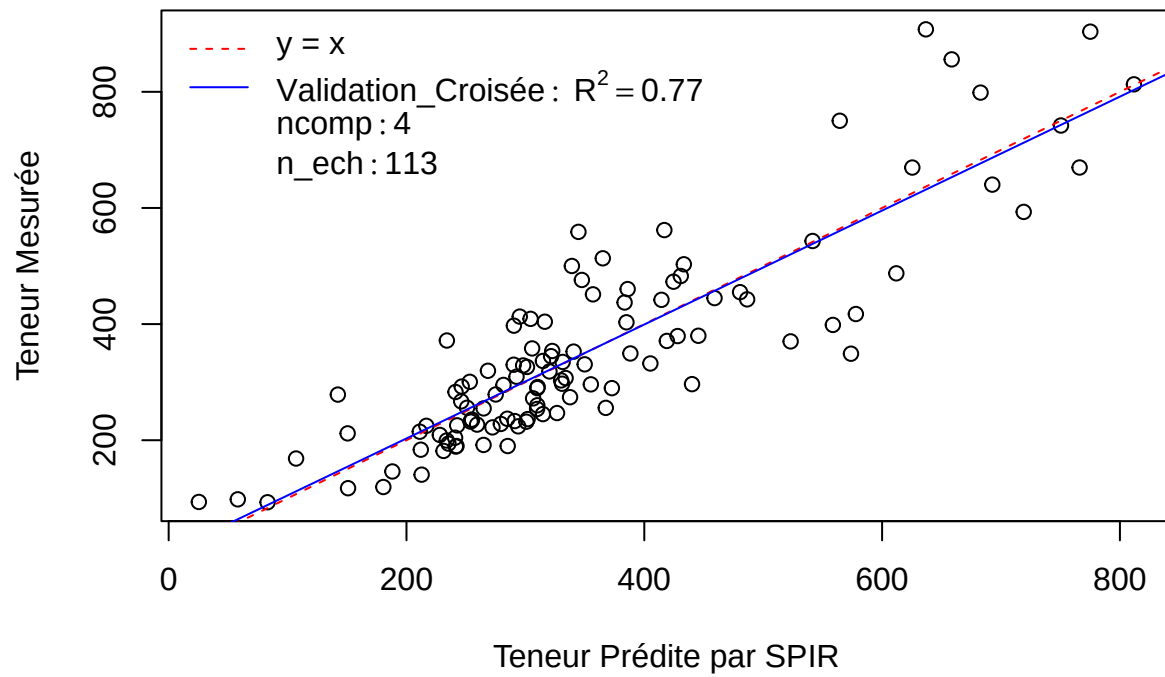
```
## [1] "C18.2"
## [1] "RANGE:1299-1750nm + SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.952"
## [1] "SEP = 0.251"
```

C18.2



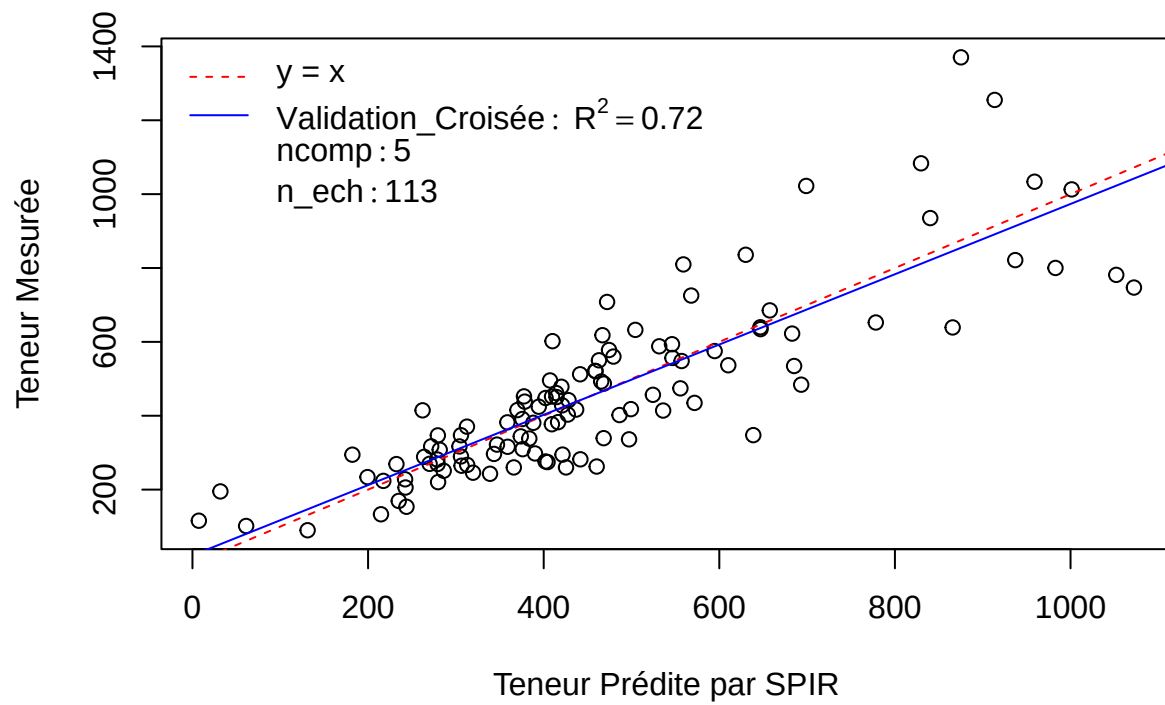
```
## [1] "trans.alpha.carotene"
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.768"
## [1] "SEP = 82.818"
```

trans.alpha.carotene



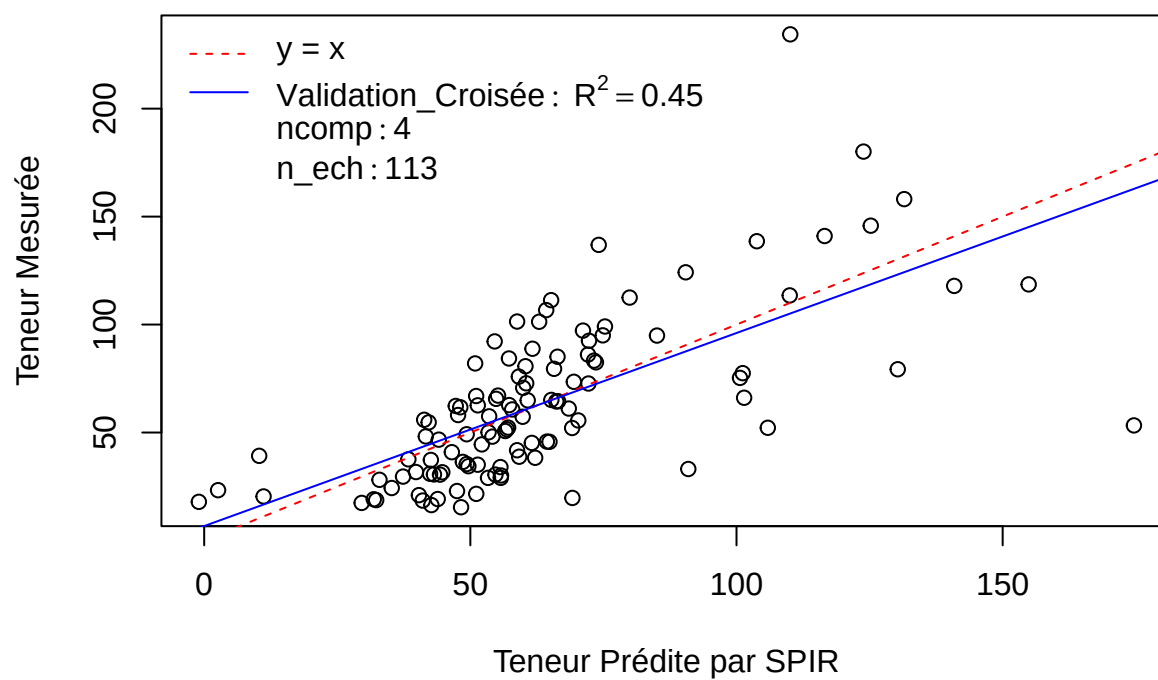
```
## [1] "trans.beta.carotene"  
## [1] "SNV + DERIV.2nde (poly=3;window=3)"  
## [1] "R2 = 0.724"  
## [1] "SEP = 124.529"
```


trans.beta.carotene



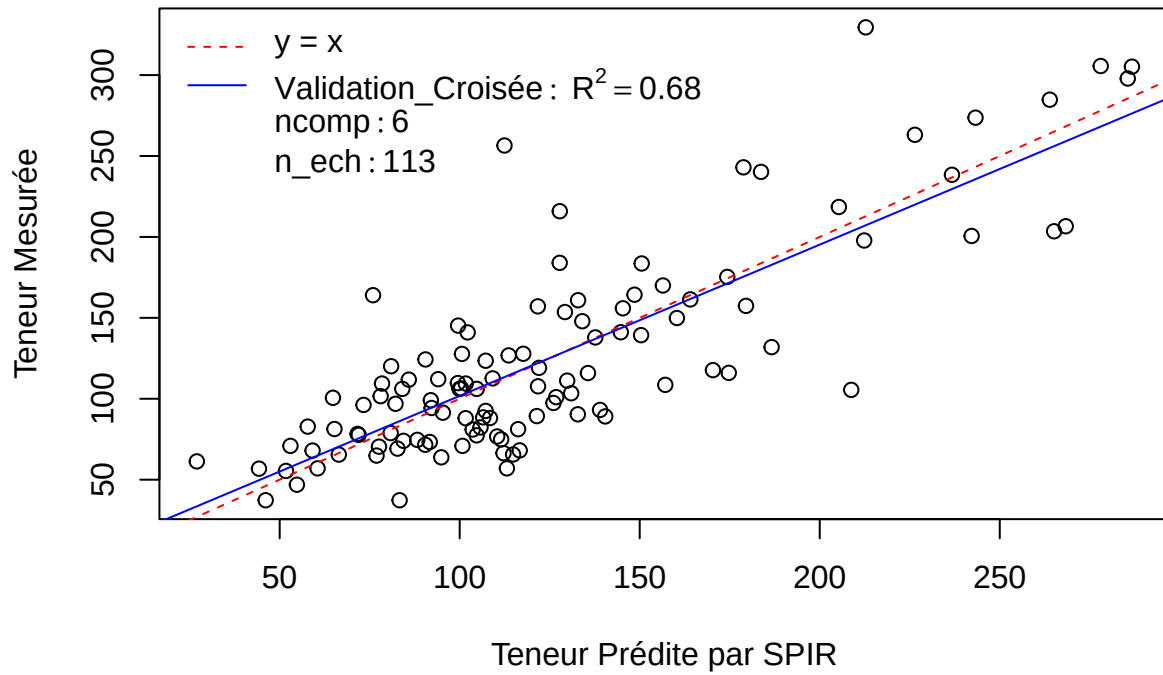
```
## [1] "X13.cis.beta.carotene"  
## [1] "SNV + DERIV.2nde (poly=3;window=3)"  
## [1] "R2 = 0.453"  
## [1] "SEP = 28.154"
```

X13.cis.beta.carotene

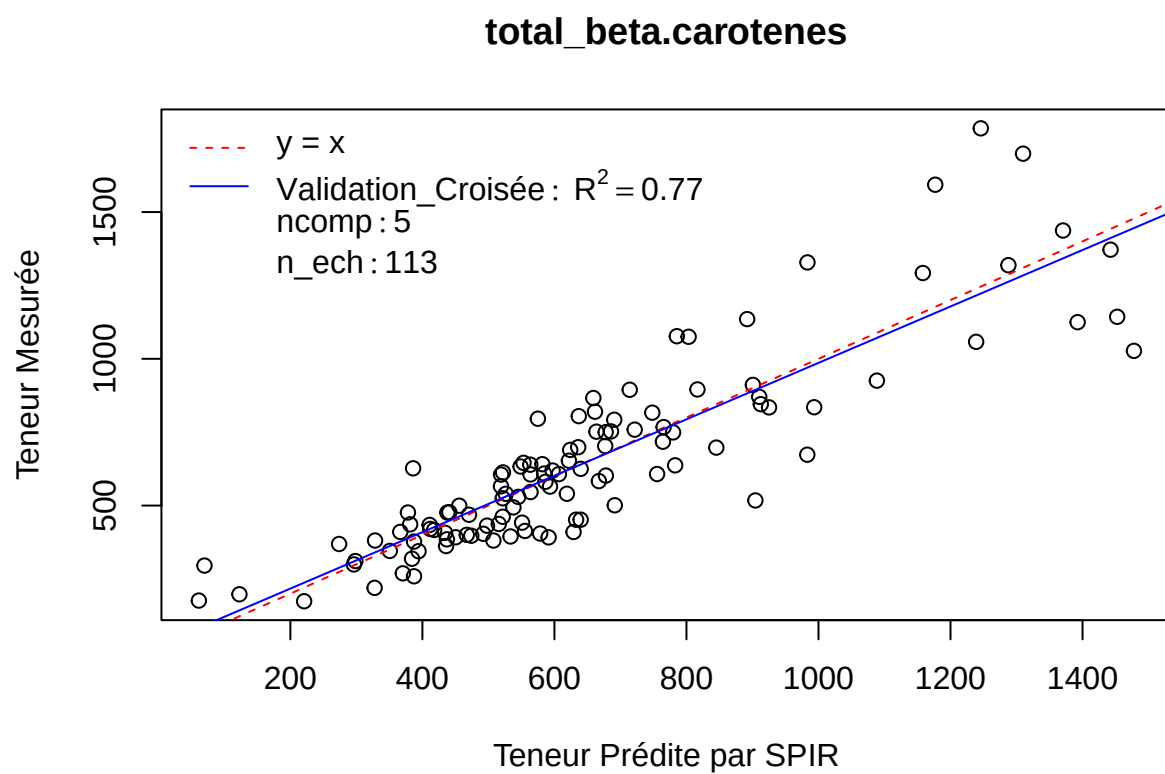


```
## [1] "X9.cis.beta.carotene"
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.678"
## [1] "SEP = 36.435"
```

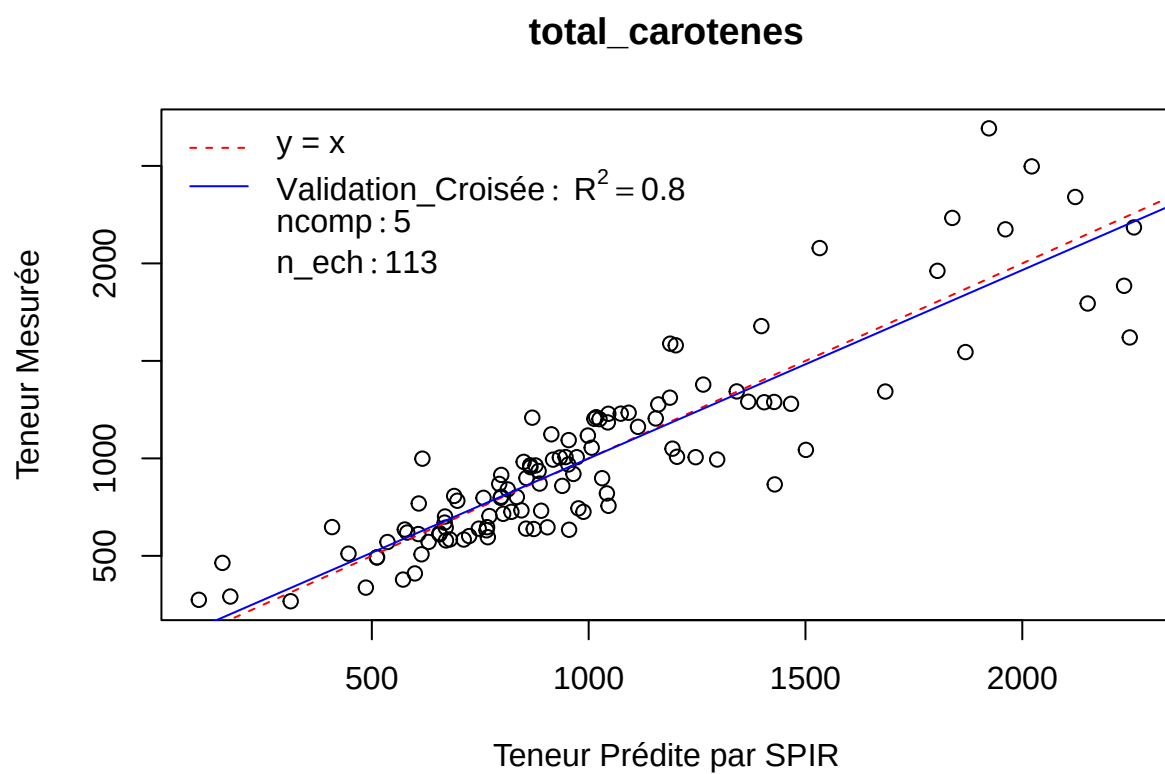
X9.cis.beta.carotene



```
## [1] "total_beta.carotenes"  
## [1] "SNV + DERIV.2nde (poly=3;window=3)"  
## [1] "R2 = 0.774"  
## [1] "SEP = 151.729"
```

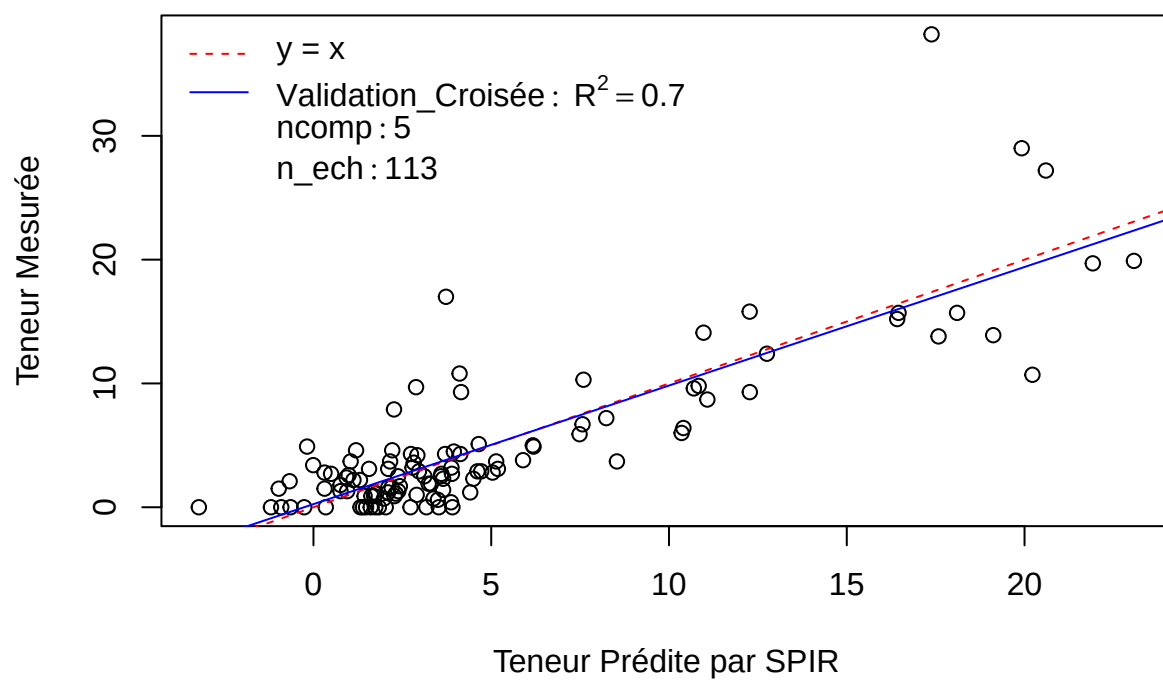


```
## [1] "total_carotenes"
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.802"
## [1] "SEP = 215.193"
```

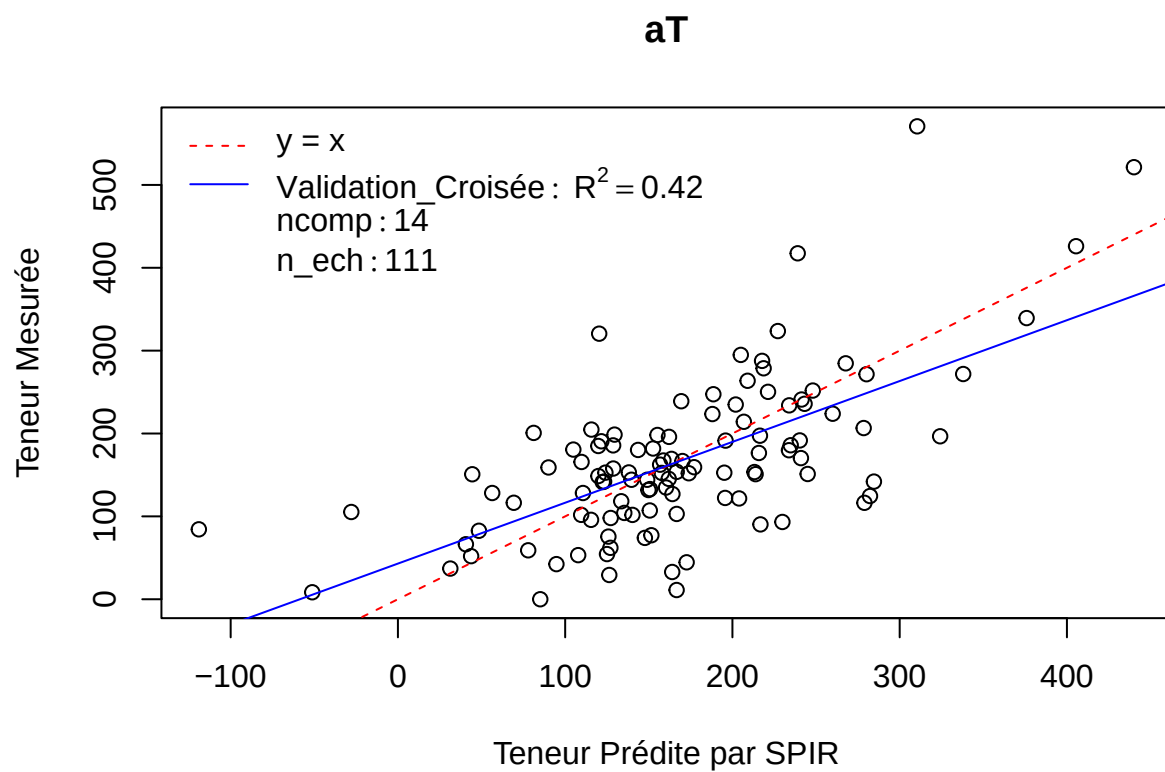


```
## [1] "Lycopene"
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.695"
## [1] "SEP = 3.558"
```

Lycopene

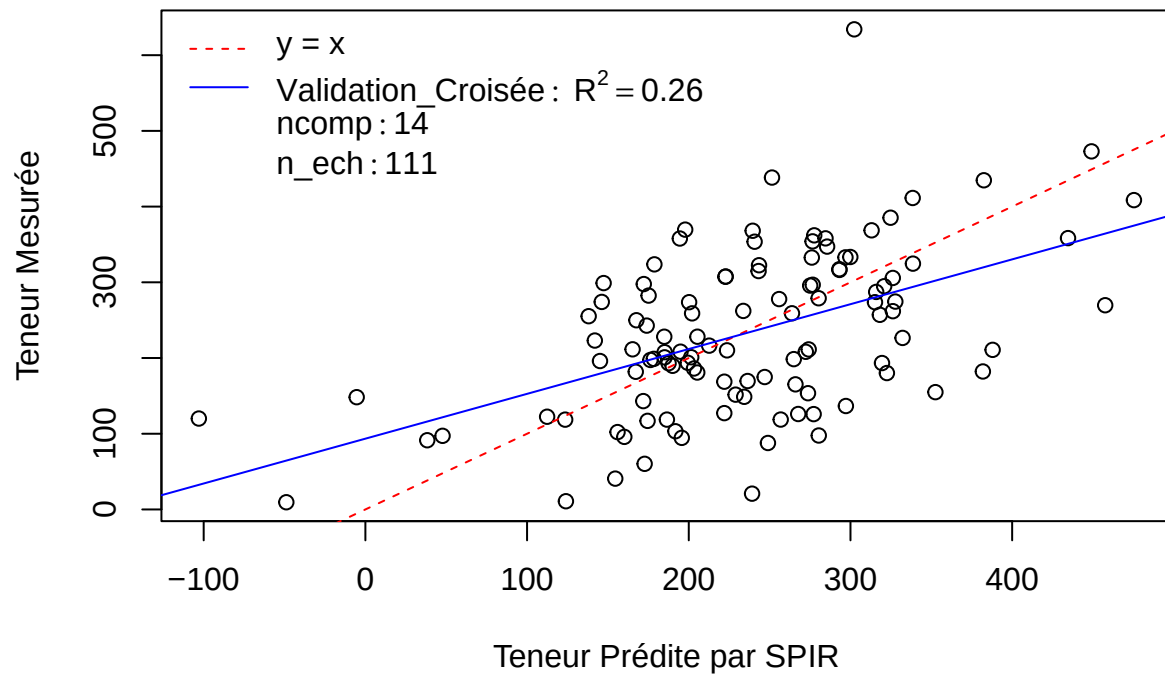


```
## [1] "aT"
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.424"
## [1] "SEP = 75.281"
```

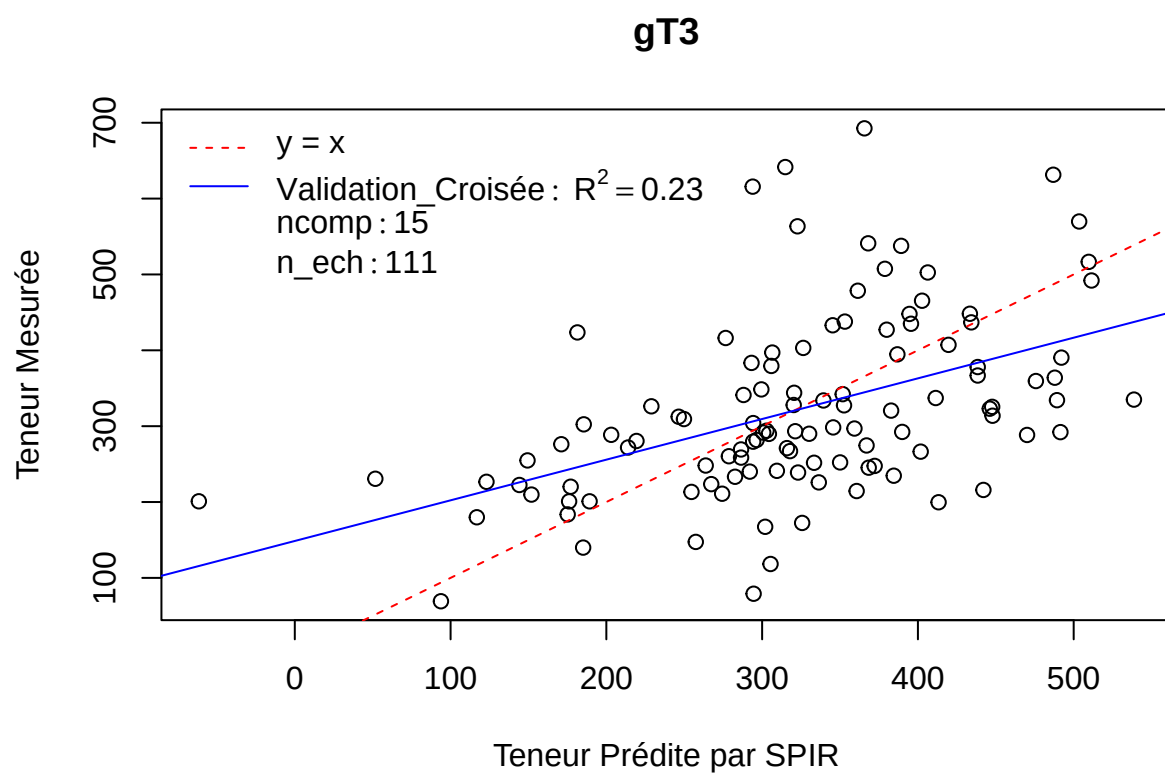


```
## [1] "aT3"
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.264"
## [1] "SEP = 98.609"
```

aT3

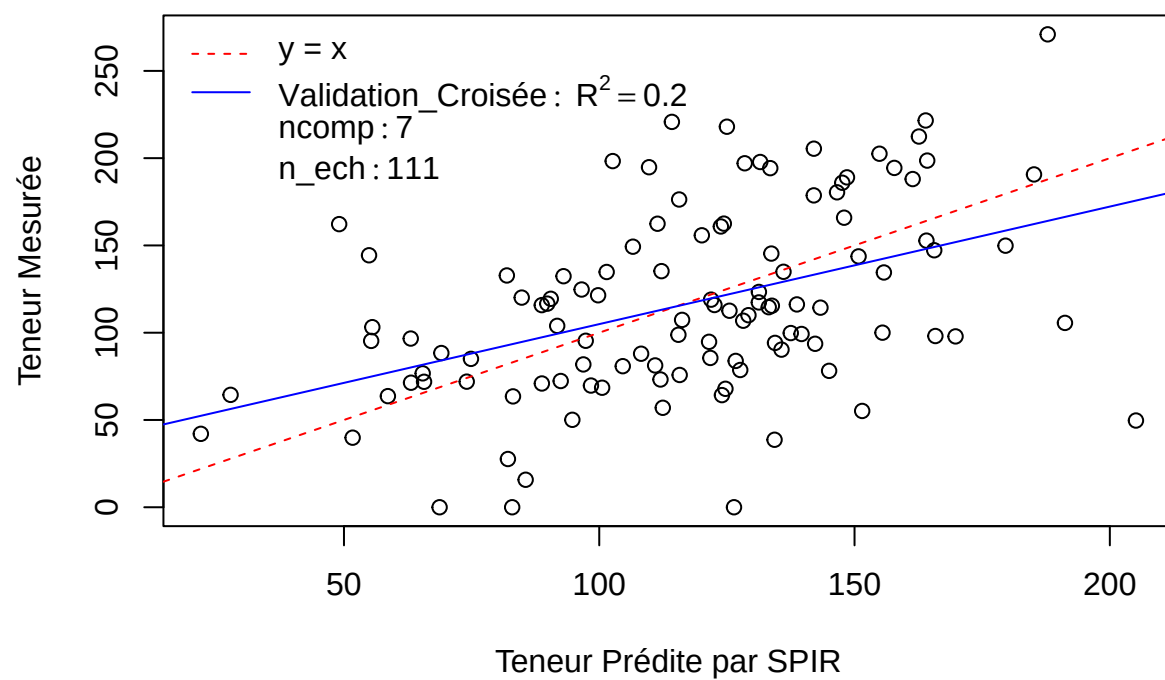


```
## [1] "gT3"  
## [1] "SNV + DERIV.2nde (poly=3;window=3)"  
## [1] "R2 = 0.228"  
## [1] "SEP = 115.61"
```

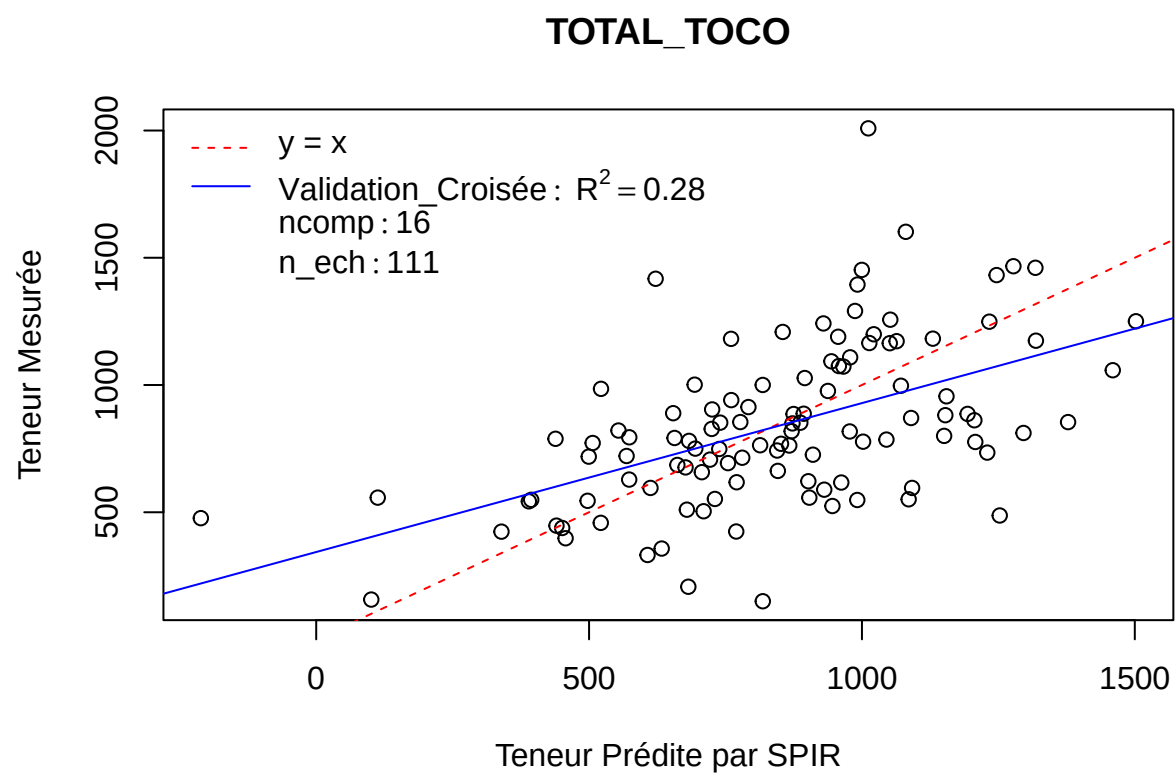



```
## [1] "dT3"
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.2"
## [1] "SEP = 49.537"
```

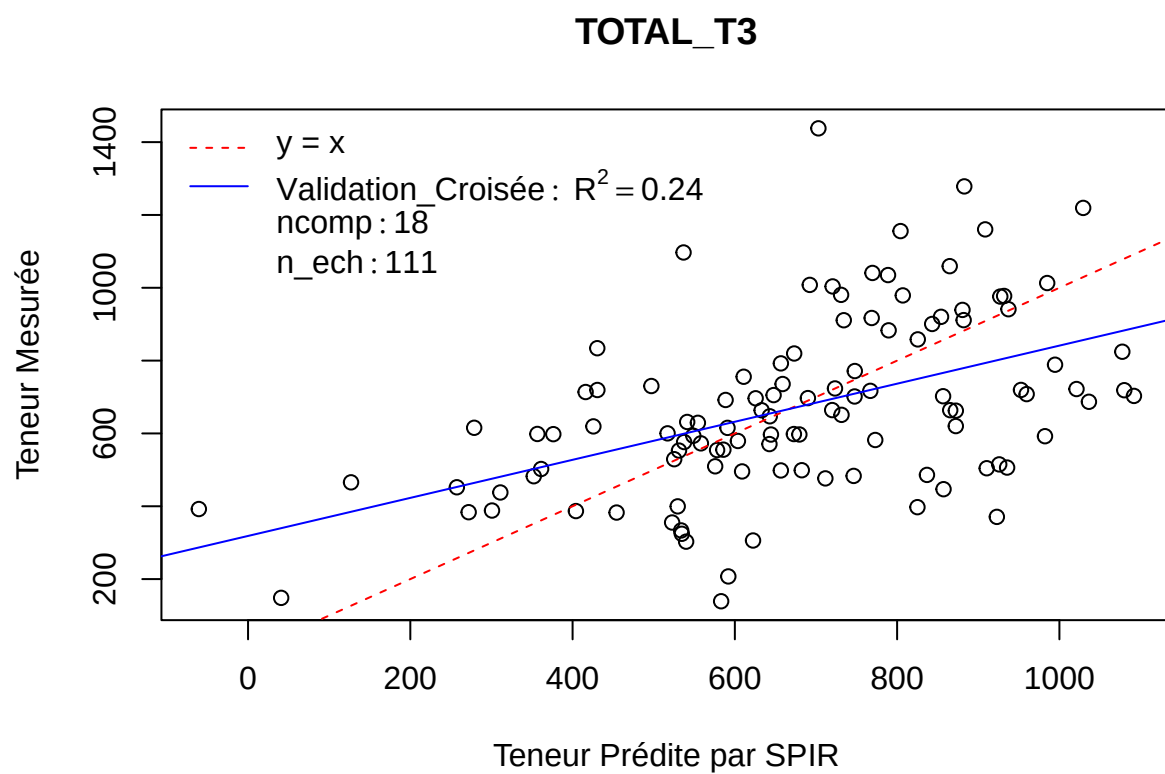
dT3



```
## [1] "TOTAL_TOCO"
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.277"
## [1] "SEP = 295.775"
```

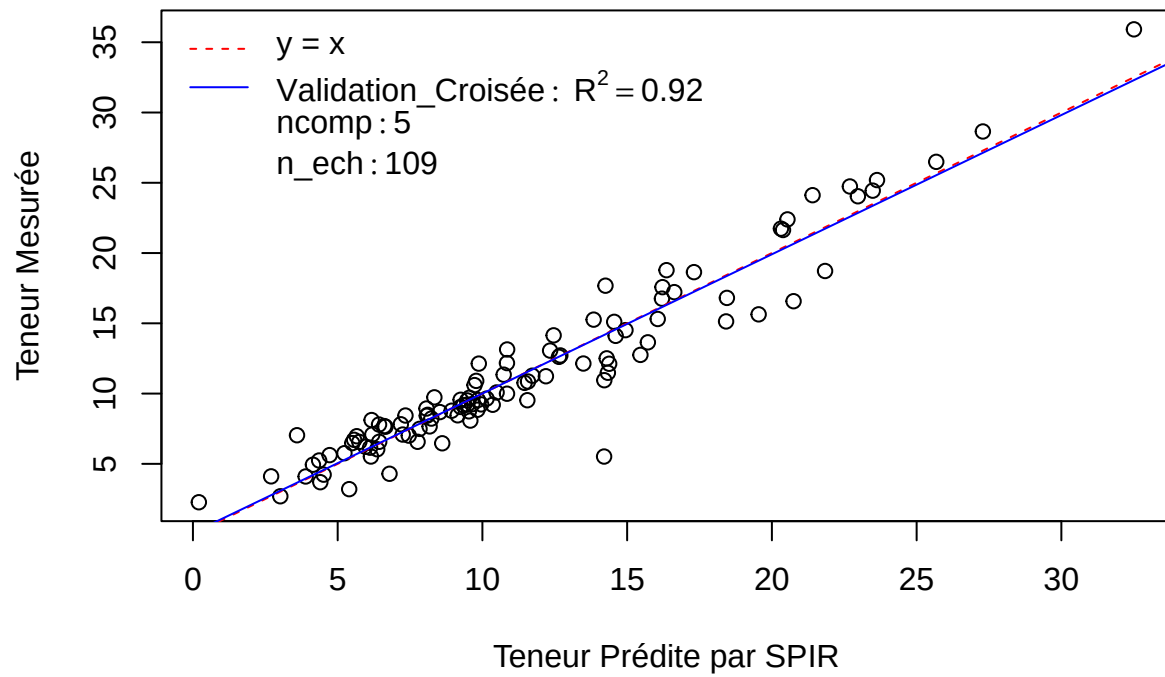


```
## [1] "TOTAL_T3"
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.237"
## [1] "SEP = 236.466"
```



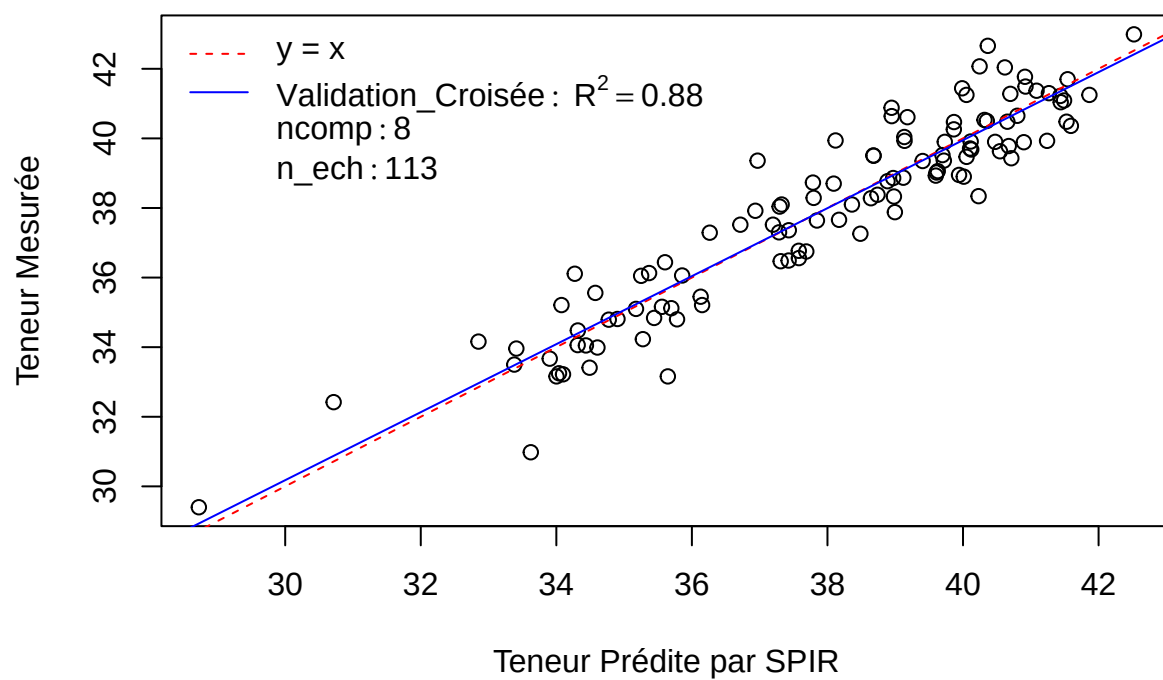
```
## [1] "FFA..."
## [1] "RANGE:1299-1750nm + SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.92"
## [1] "SEP = 1.722"
```

FFA...



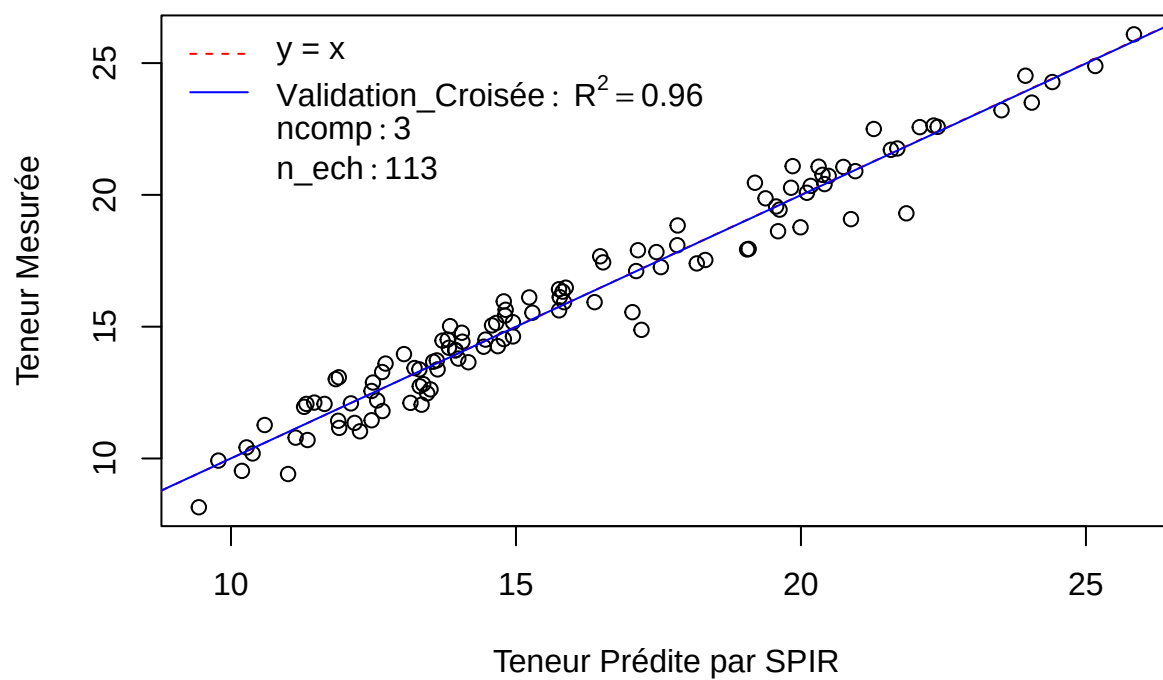
```
## [1] "L."
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.885"
## [1] "SEP = 0.952"
```

L.



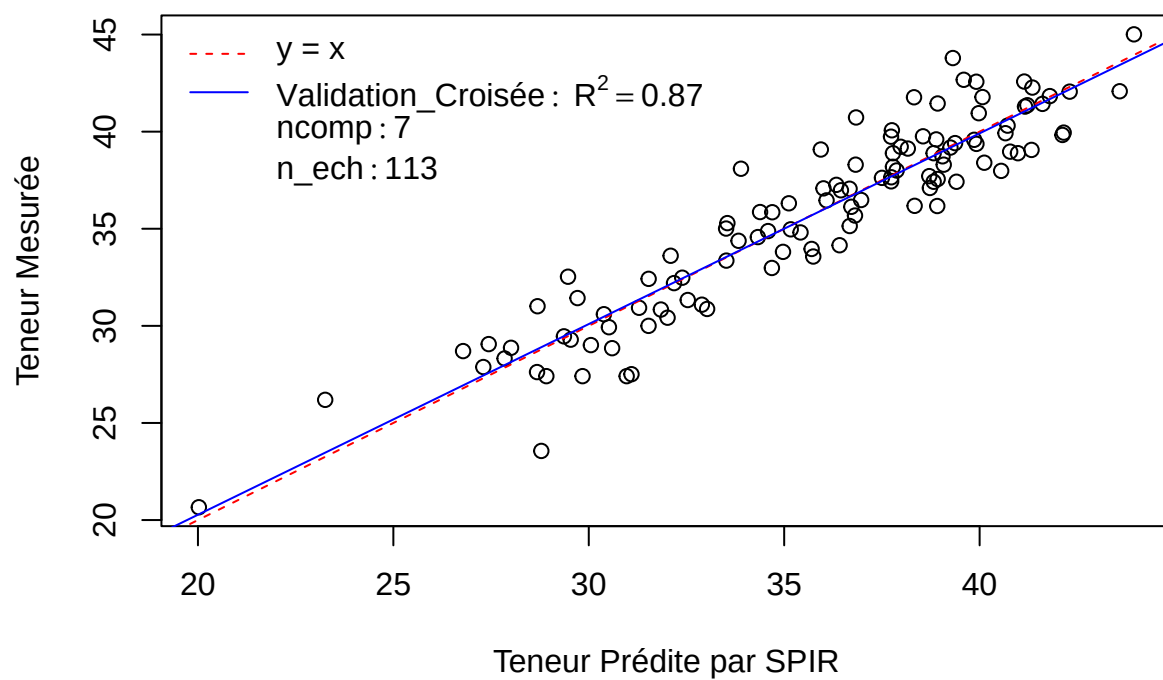
```
## [1] "a."
## [1] "RANGE:350-1200nm + SNV"
## [1] "R2 = 0.963"
## [1] "SEP = 0.768"
```

a.



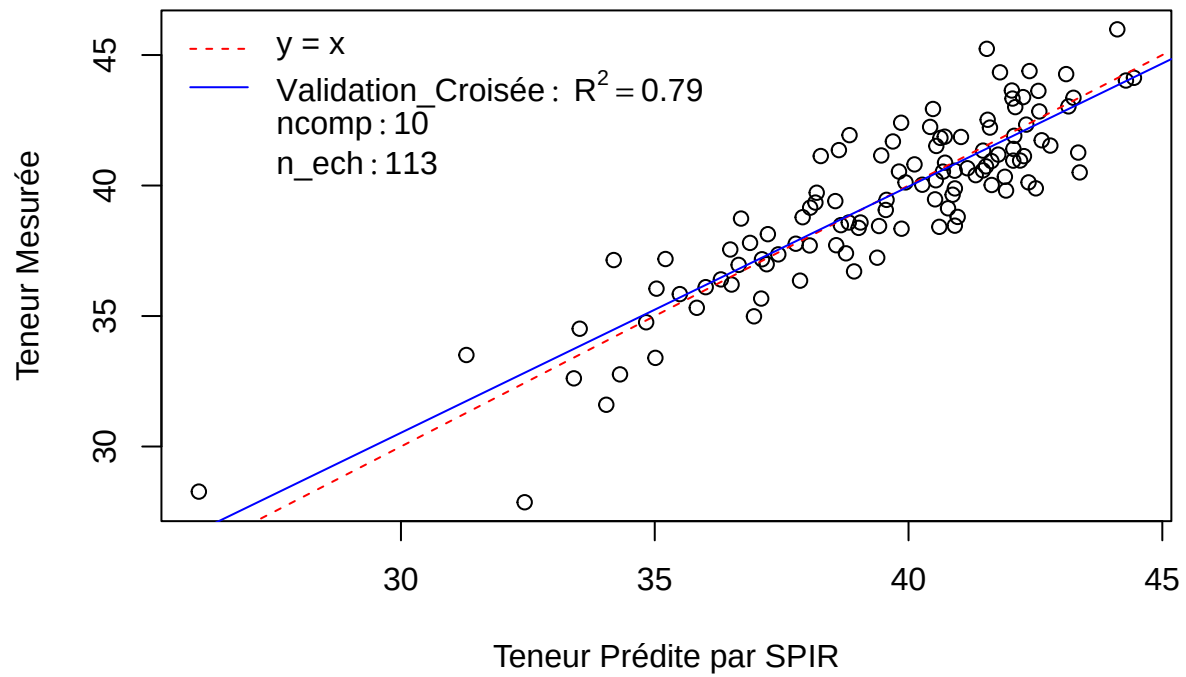
```
## [1] "b."
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.874"
## [1] "SEP = 1.729"
```

b.



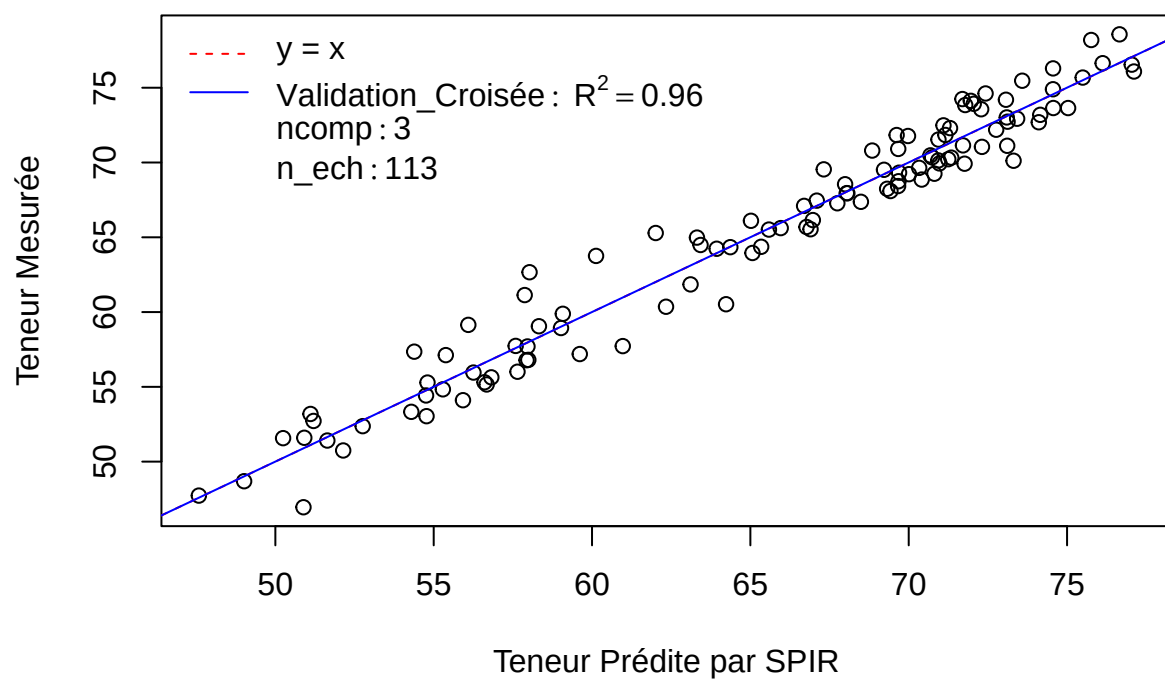
```
## [1] "C."
## [1] "SNV + DERIV.2nde (poly=3;window=3)"
## [1] "R2 = 0.789"
## [1] "SEP = 1.498"
```


C.



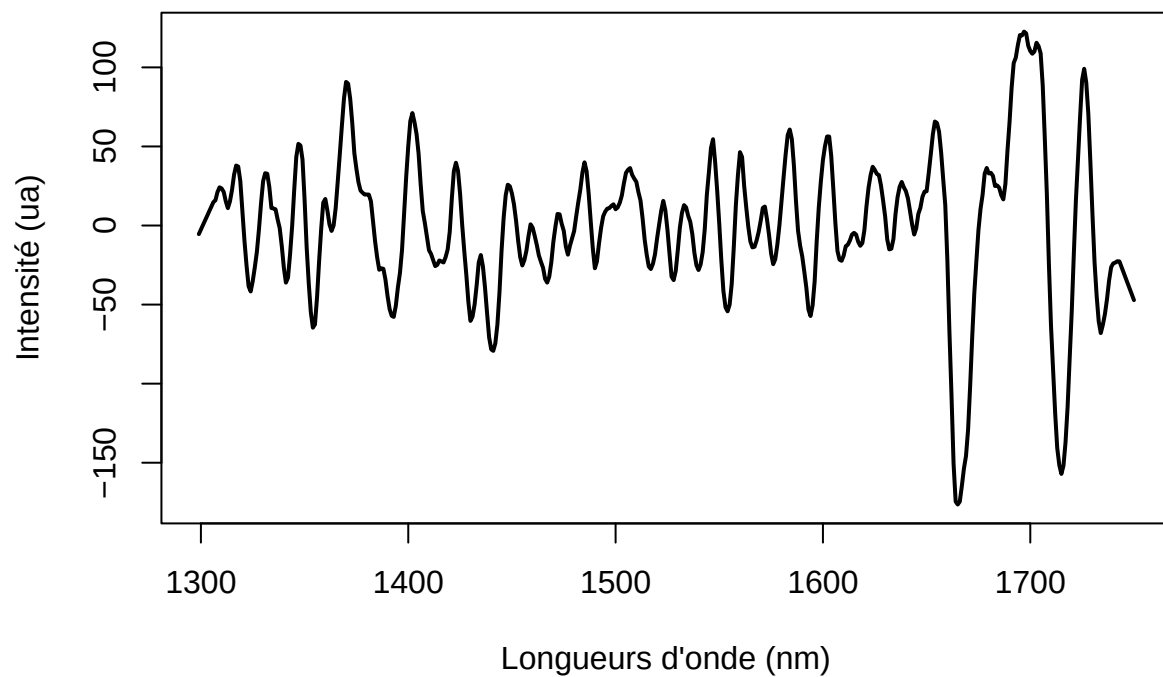
```
## [1] "h"  
## [1] "RANGE:350-1200nm + SNV"  
## [1] "R2 = 0.961"  
## [1] "SEP = 1.576"
```

h

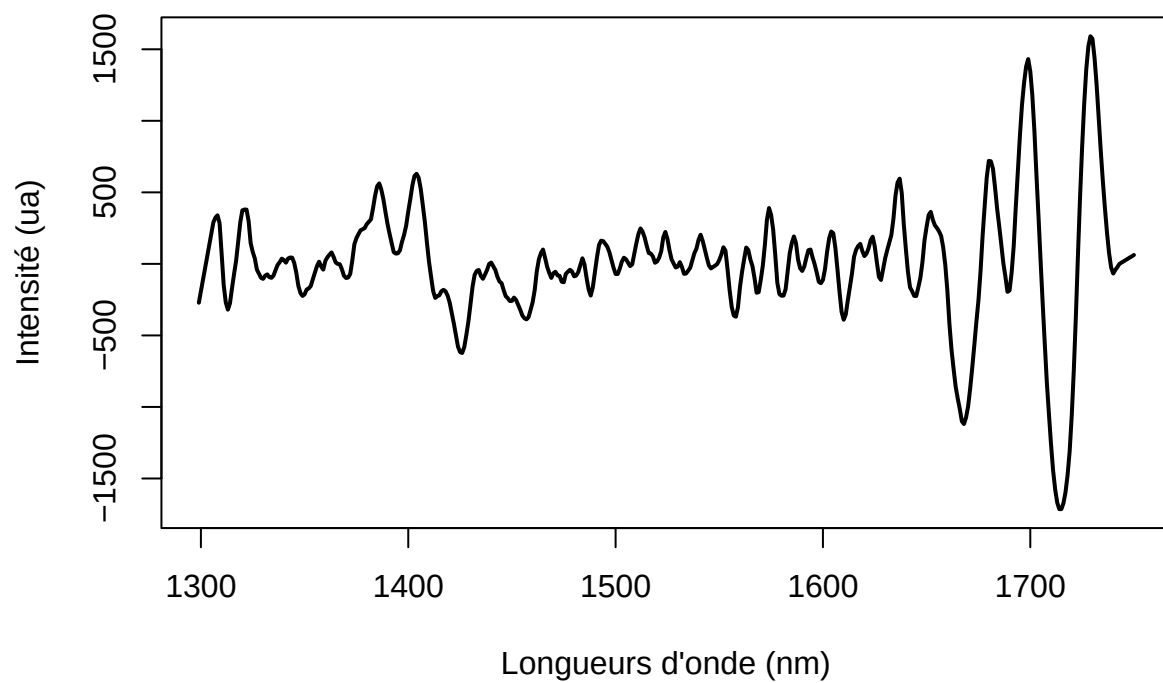


Graphique des Coefficients Beta

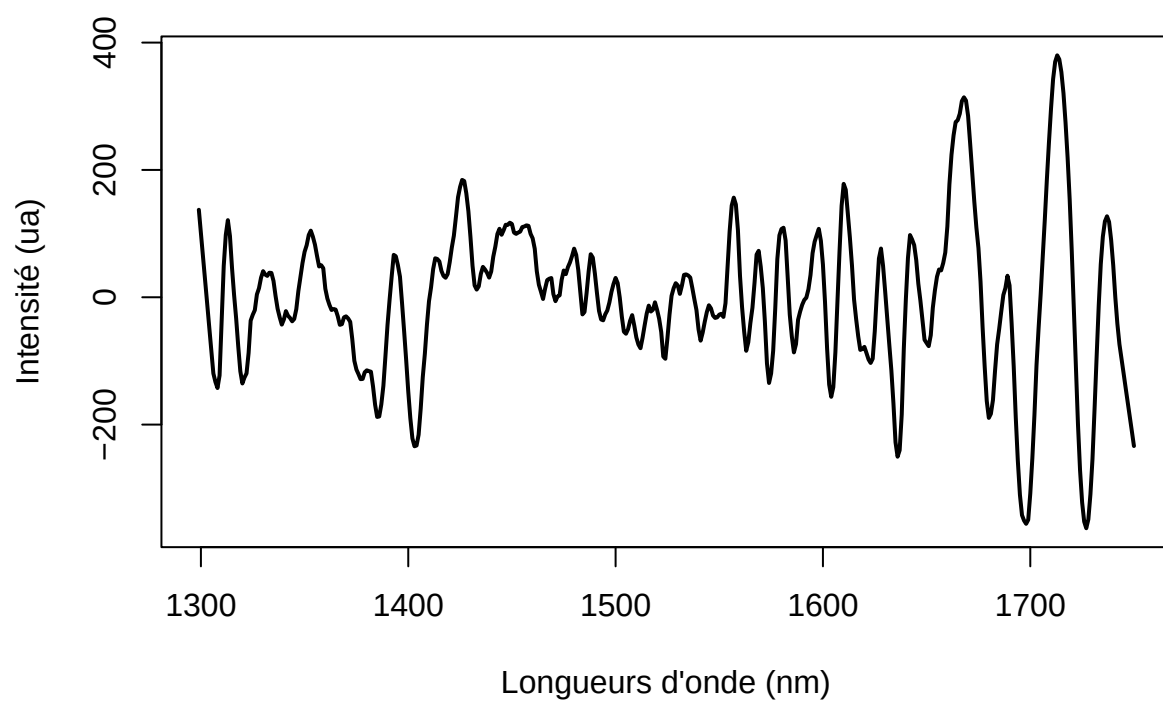
C14.0



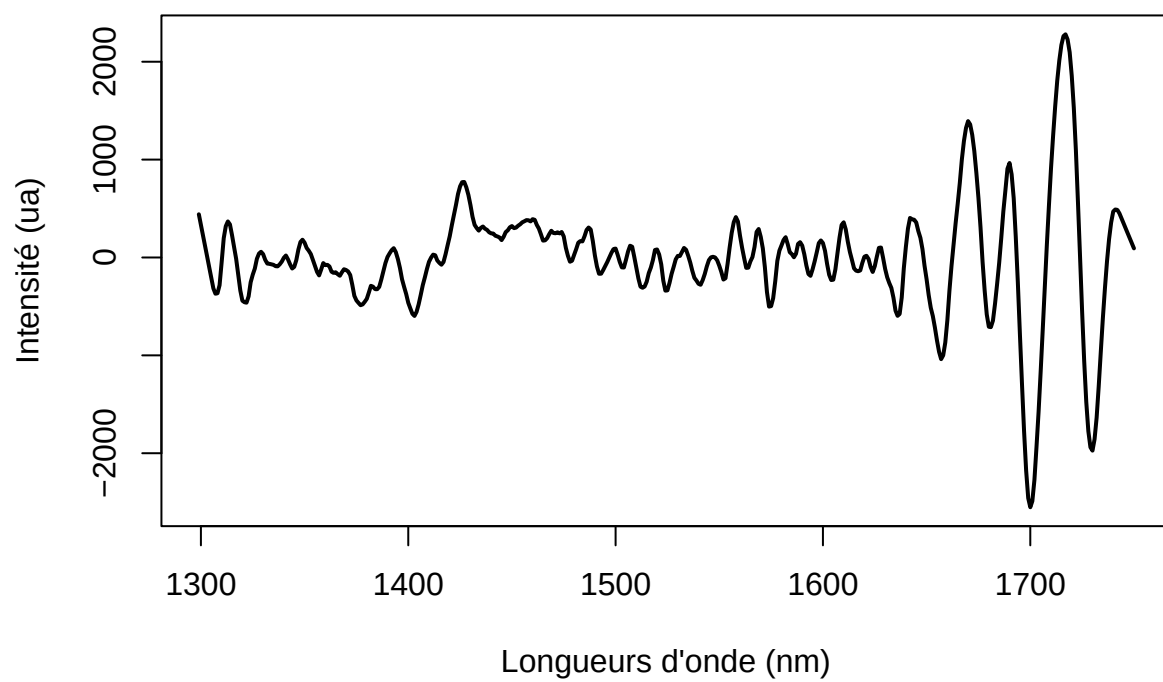
C16.0



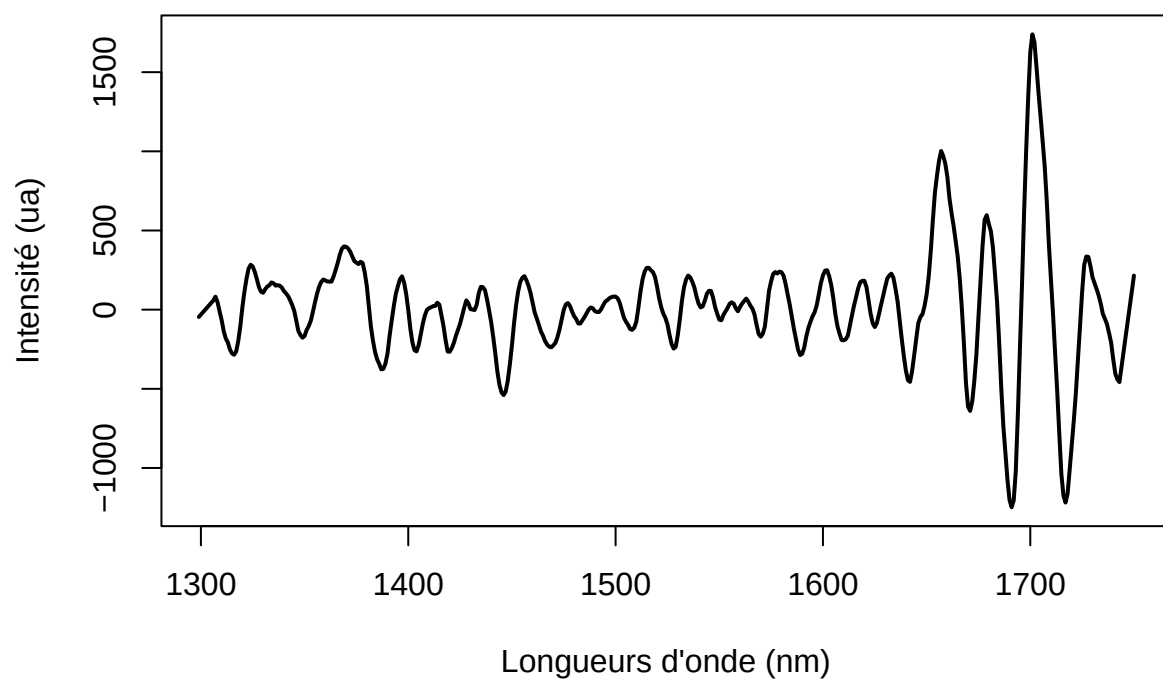
C18.0



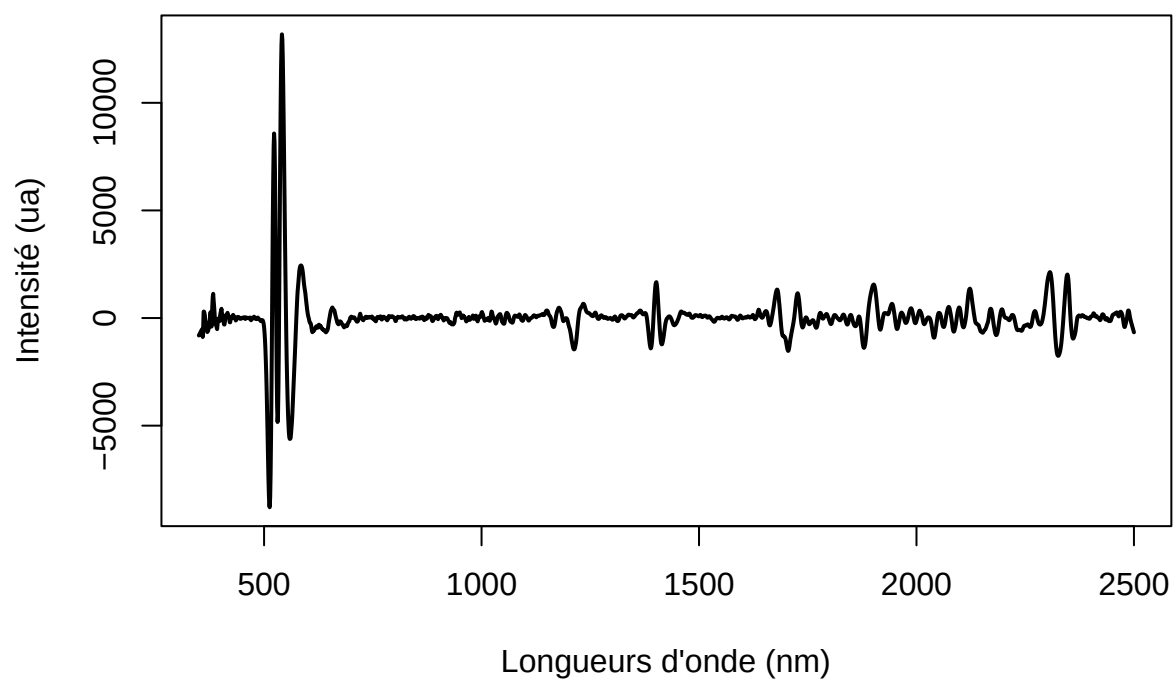
C18.1n9



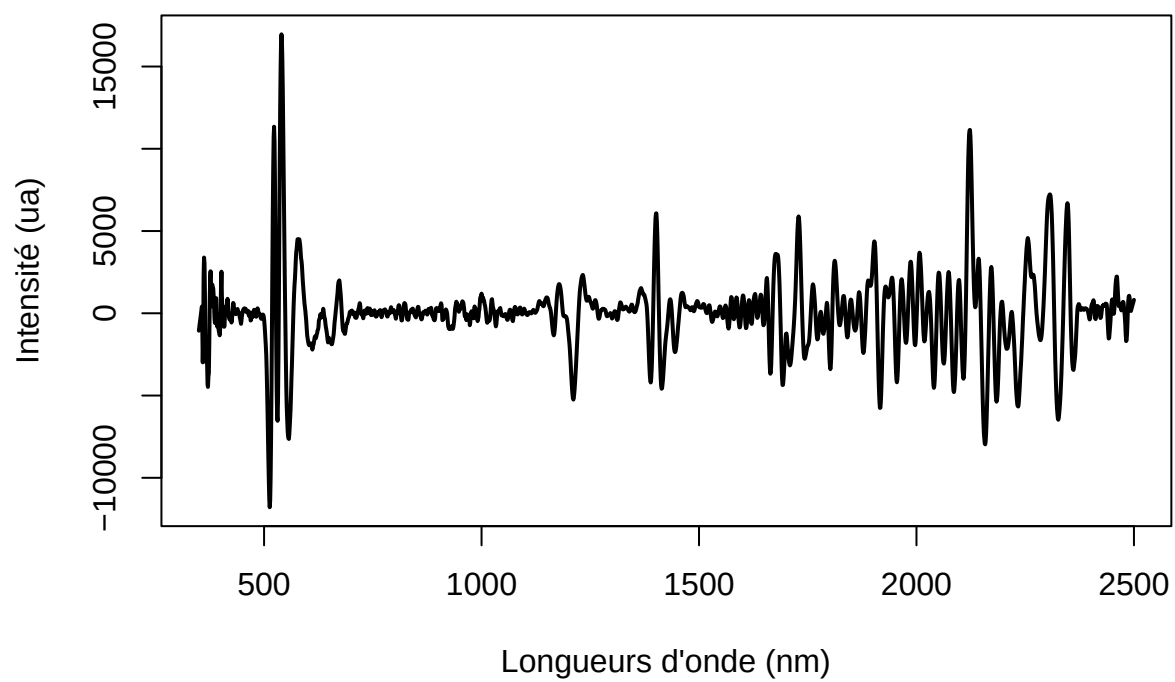
C18.2



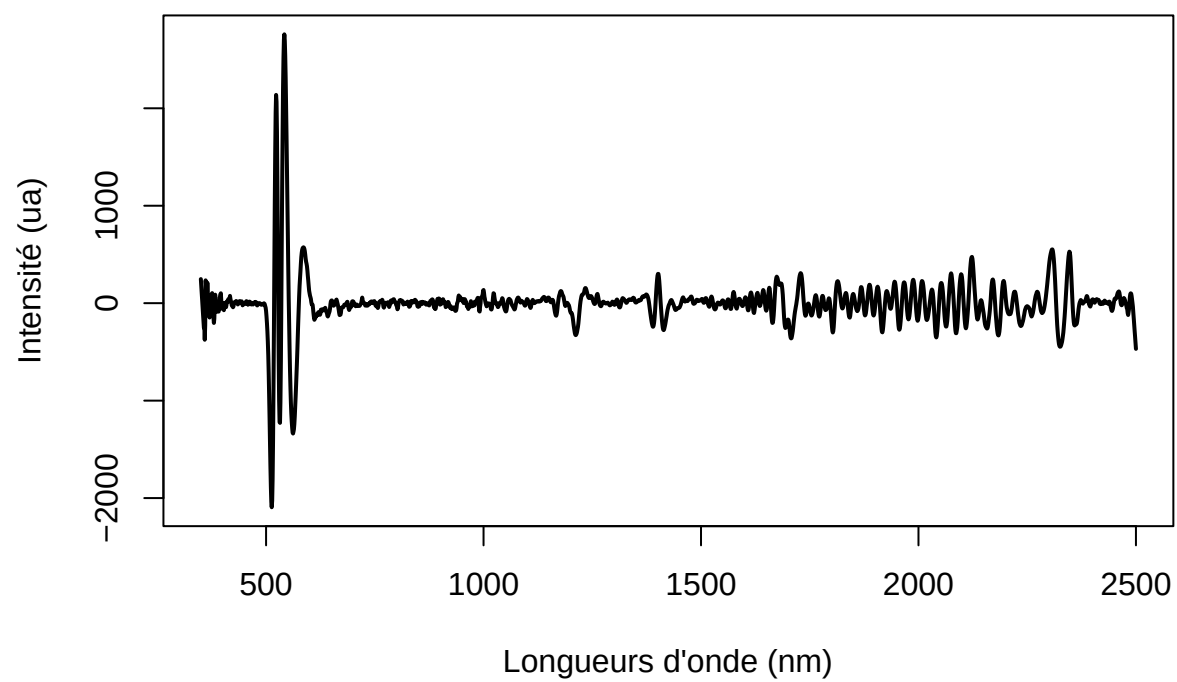
trans.alpha.carotene



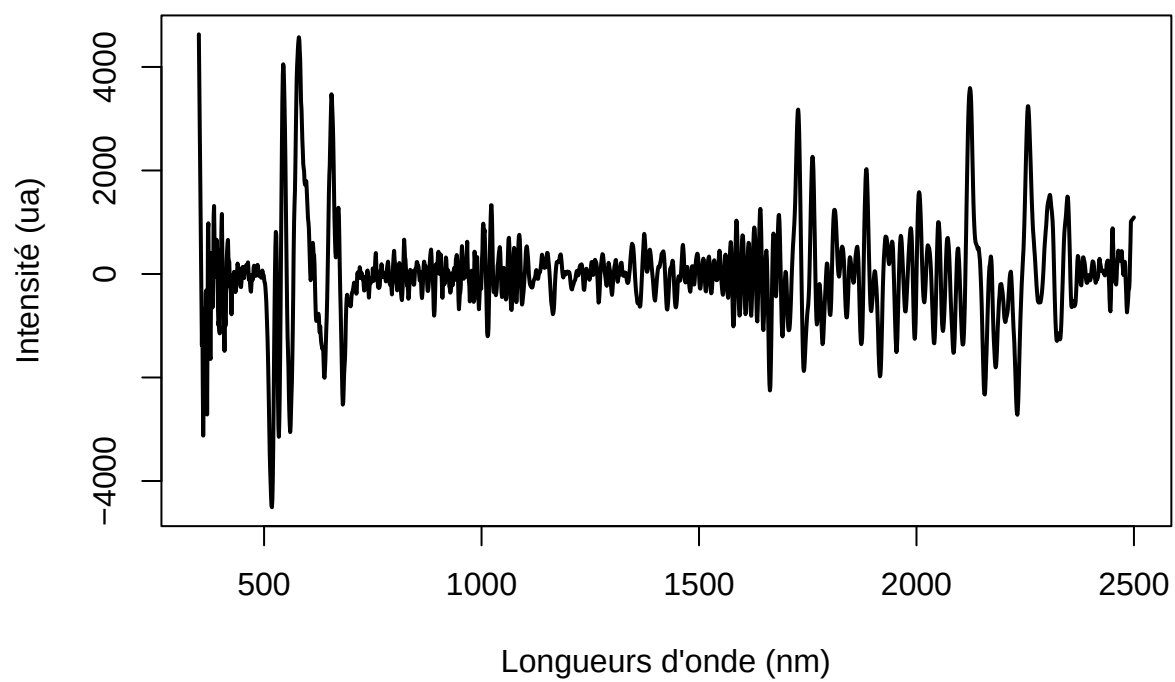
trans.beta.carotene



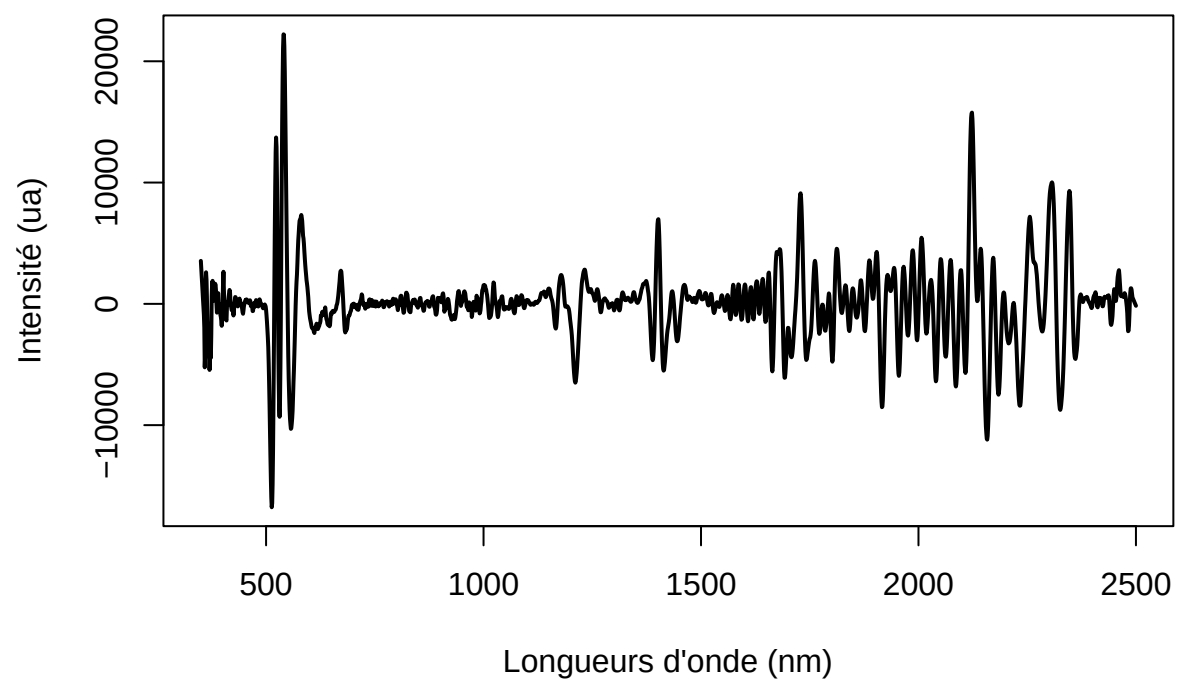
X13.cis.beta.carotene



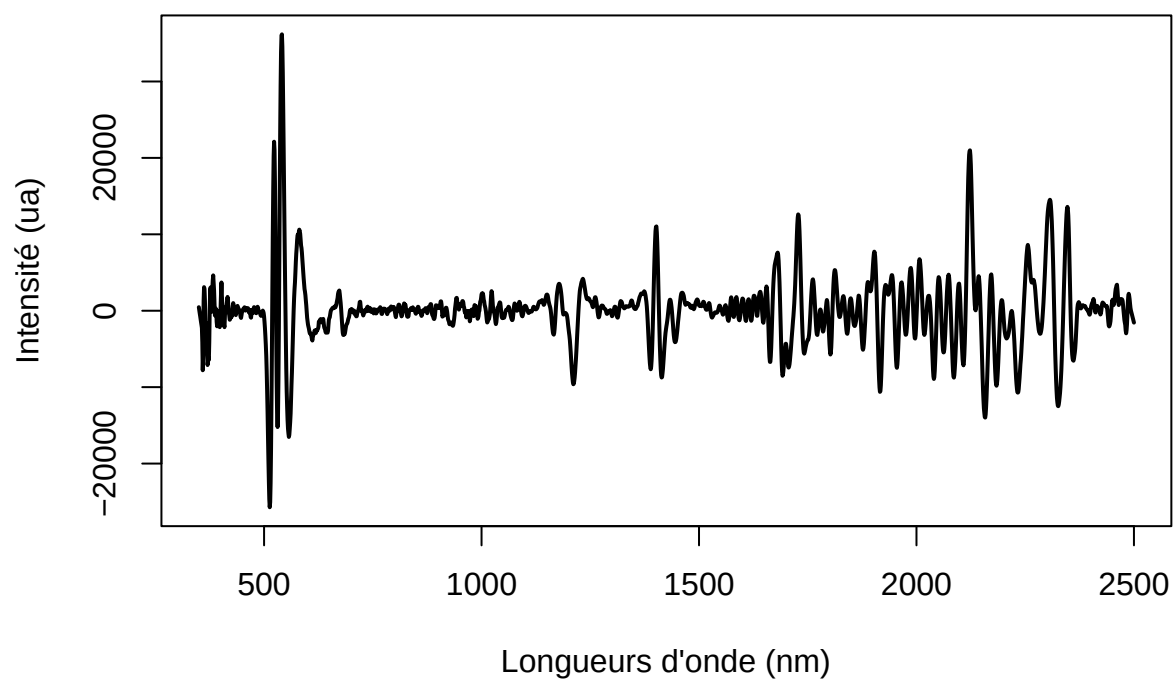
X9.cis.beta.carotene



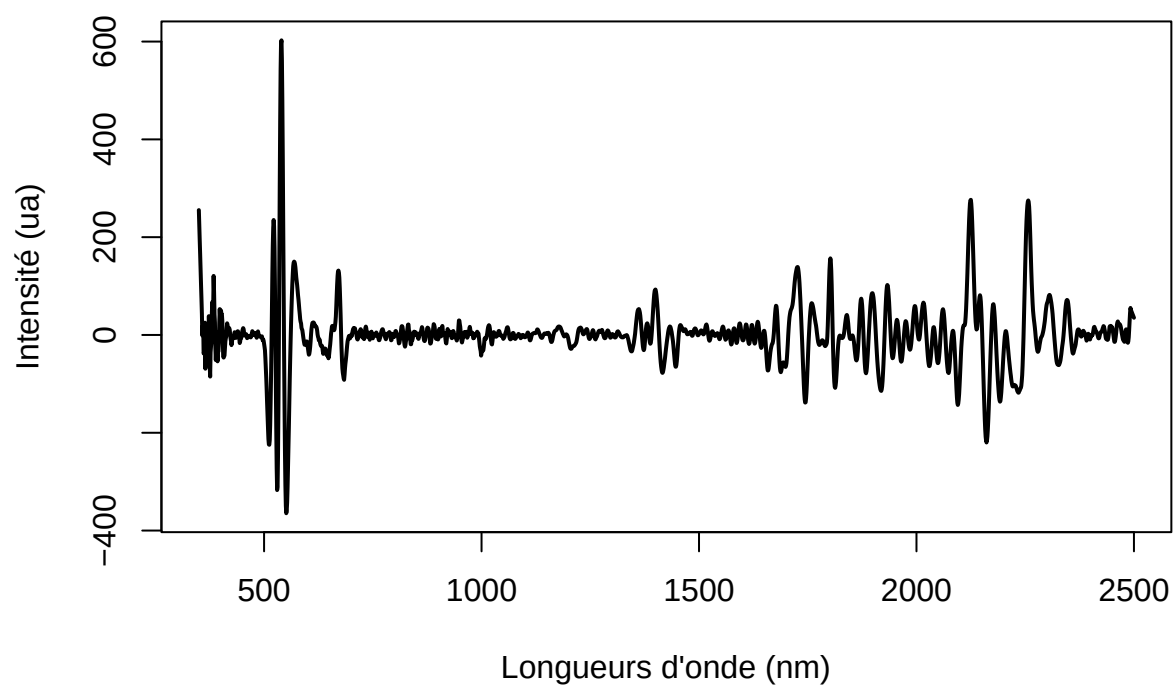
total_beta.carotenes



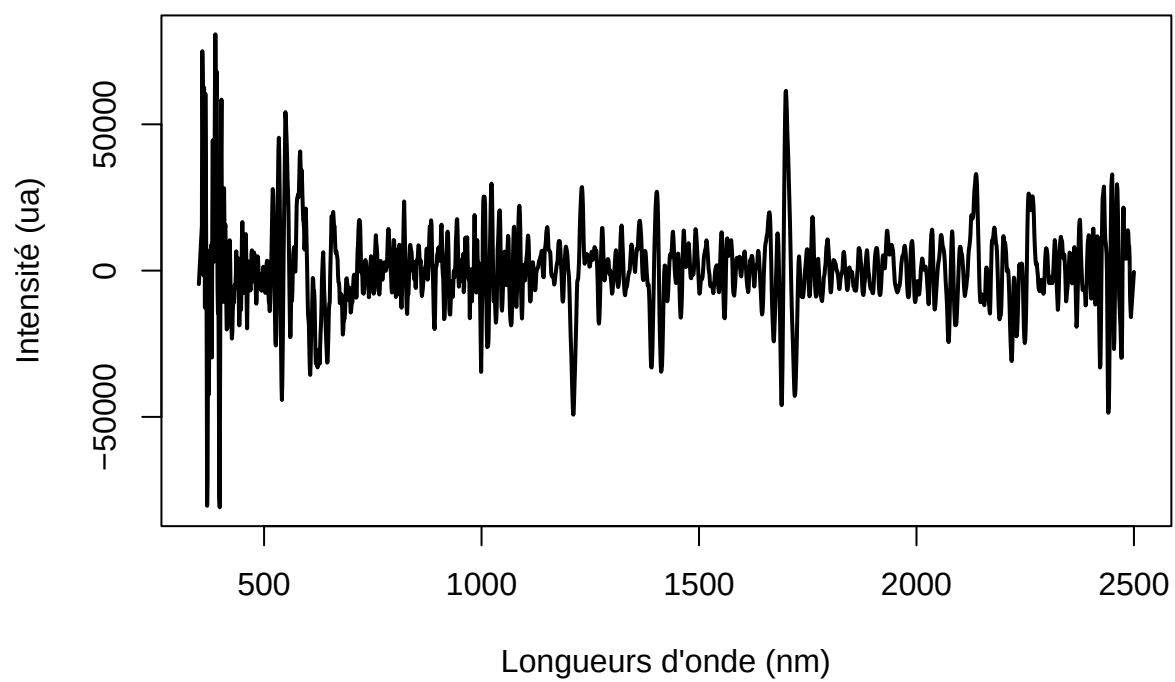
total_carotenes

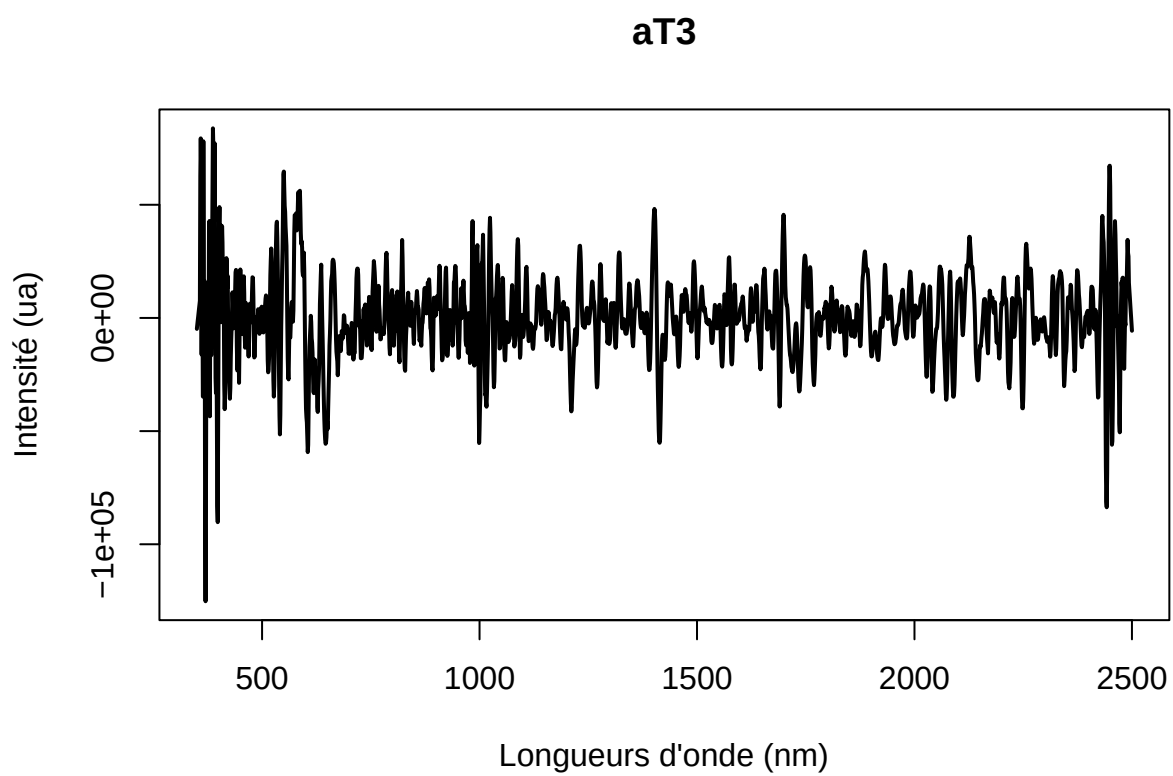


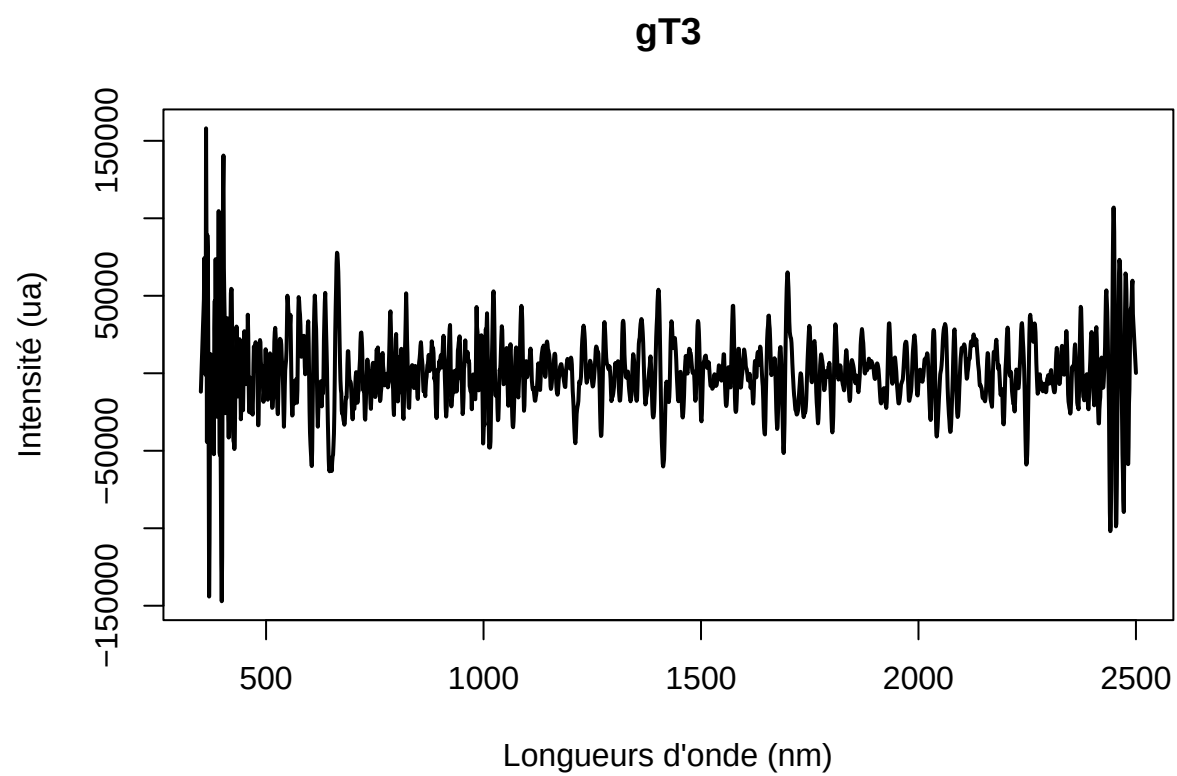
Lycopene

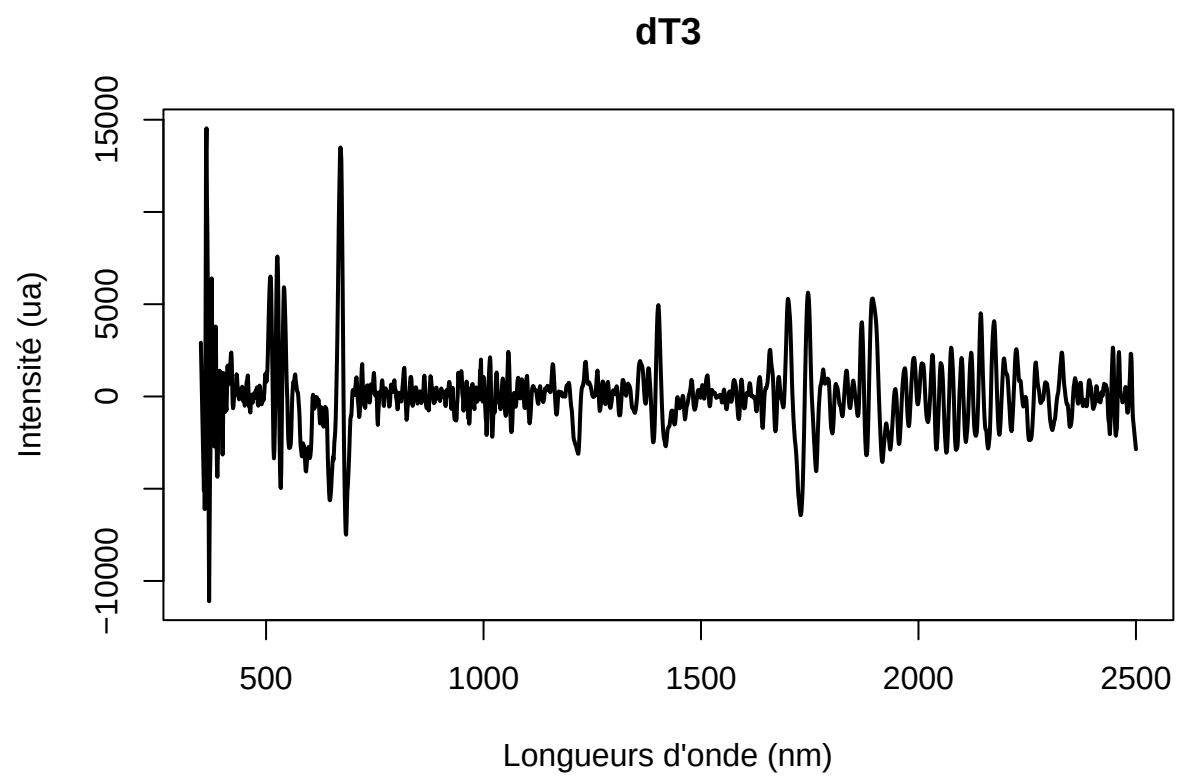


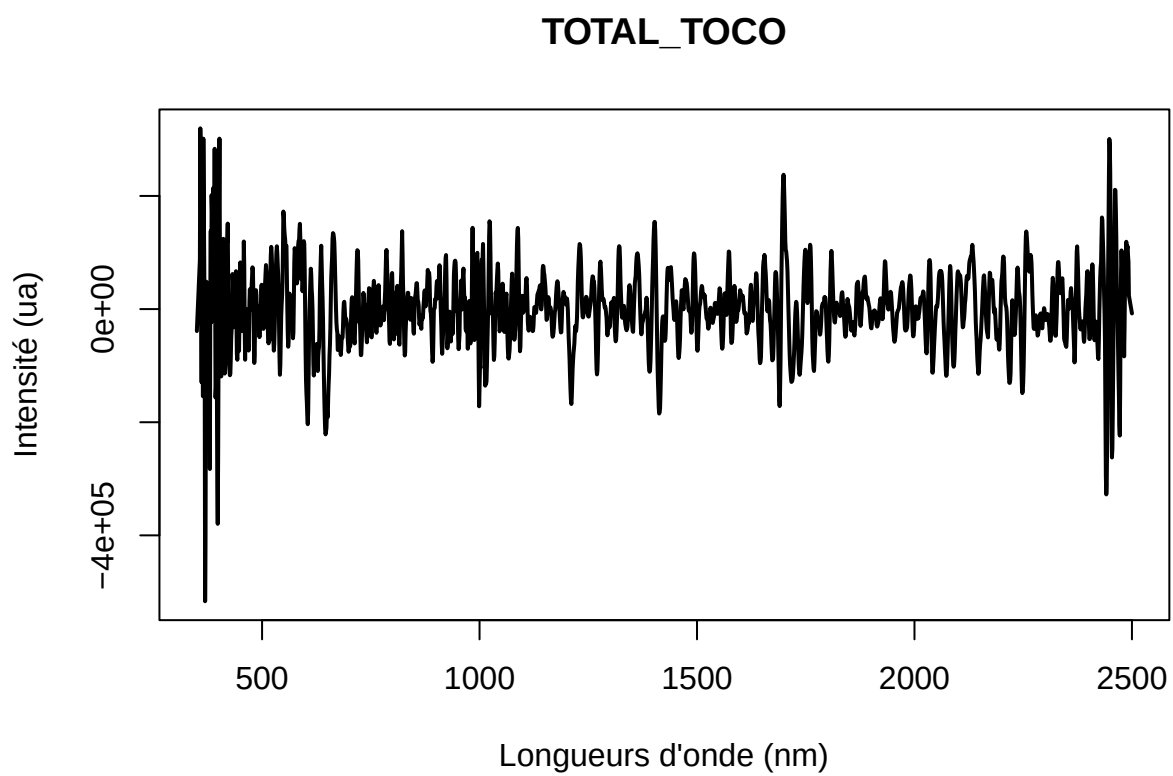
aT

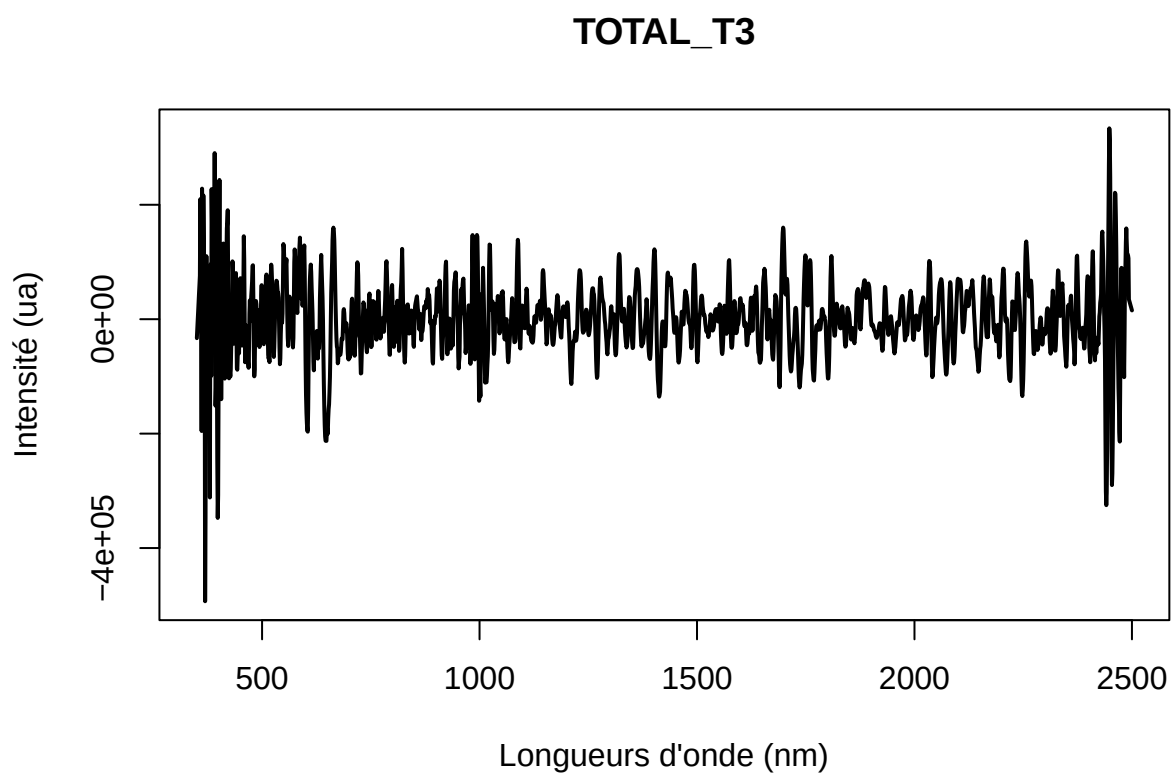




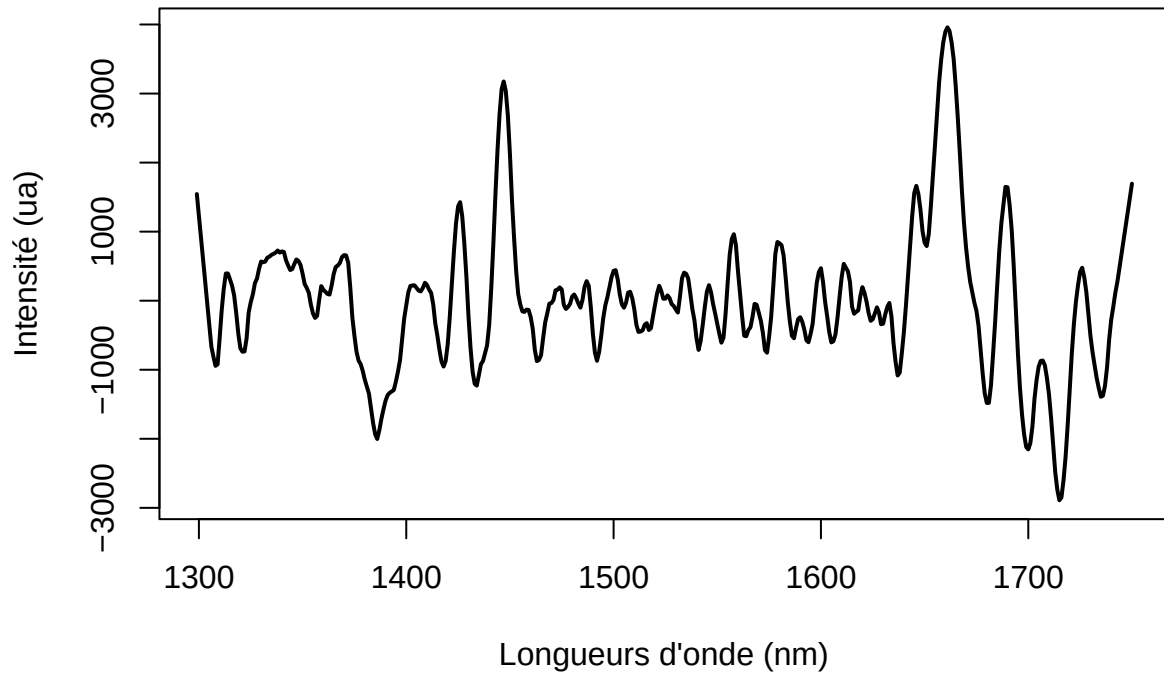




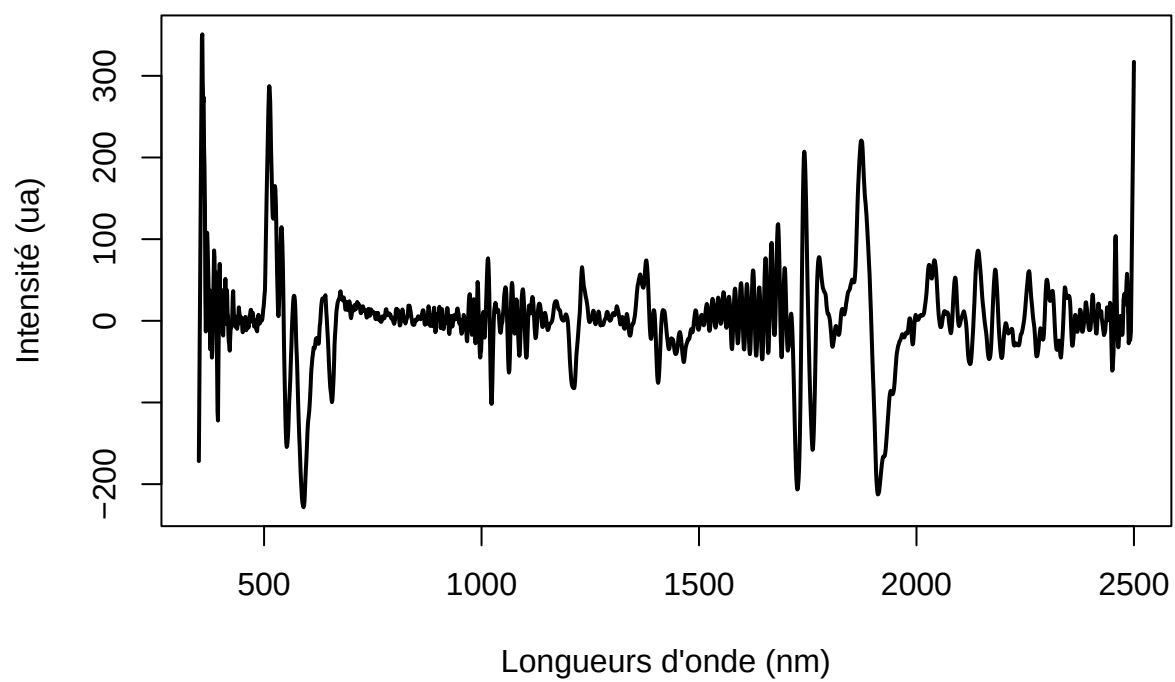




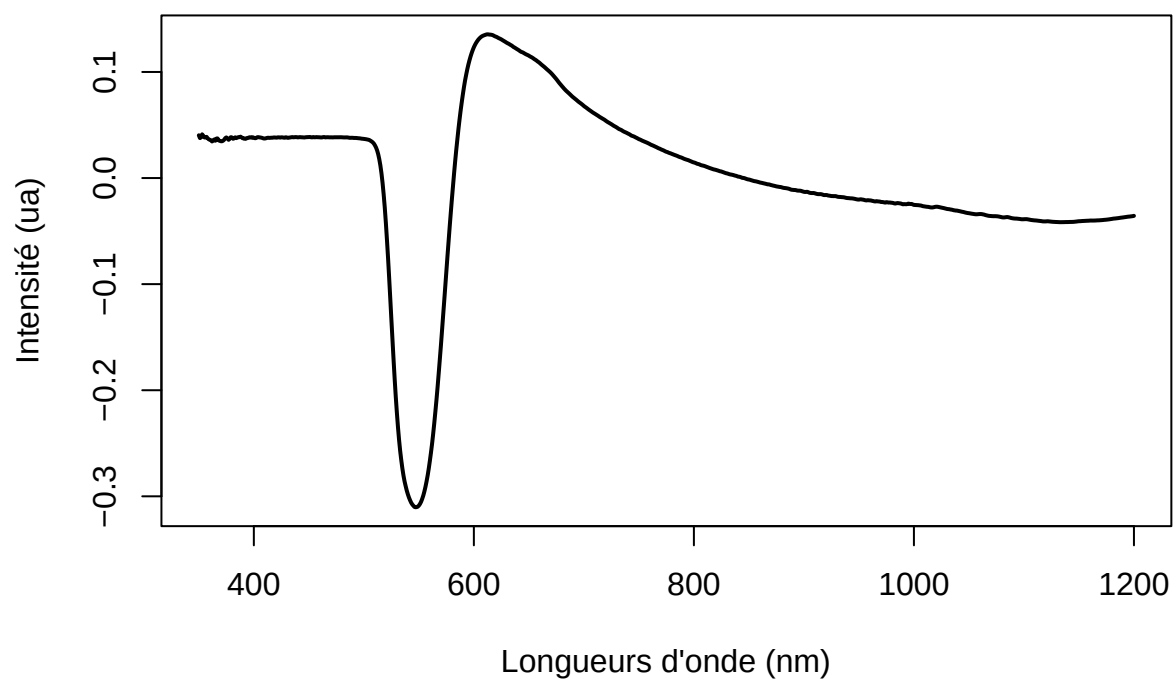
FFA...



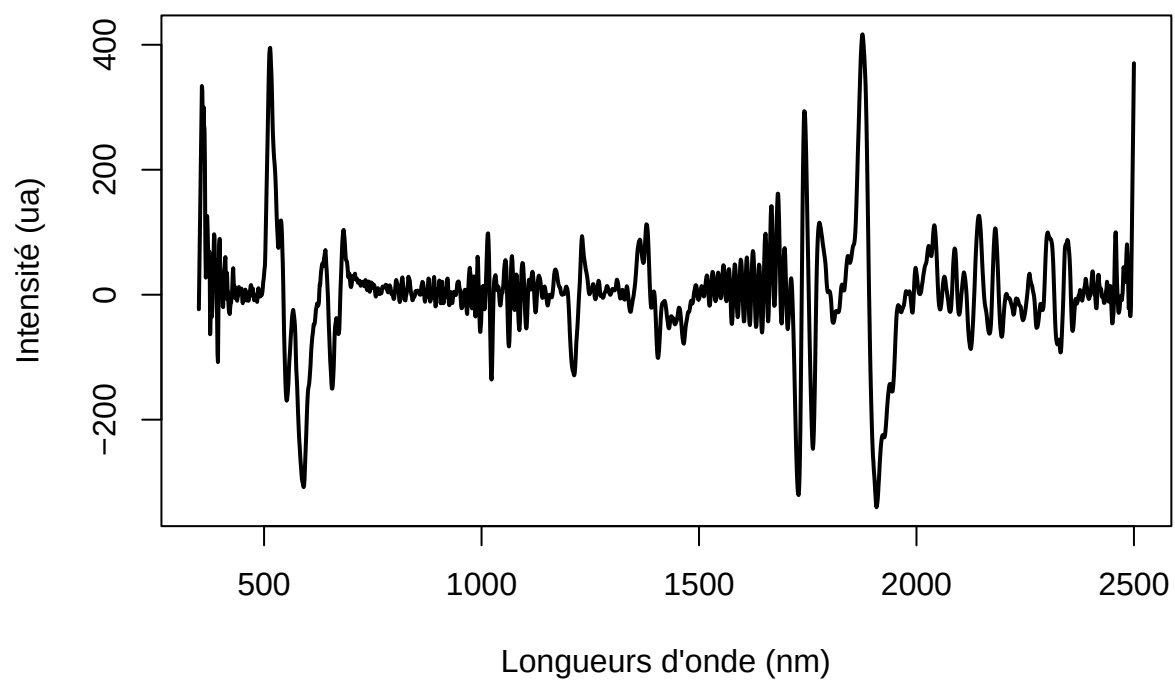
L.



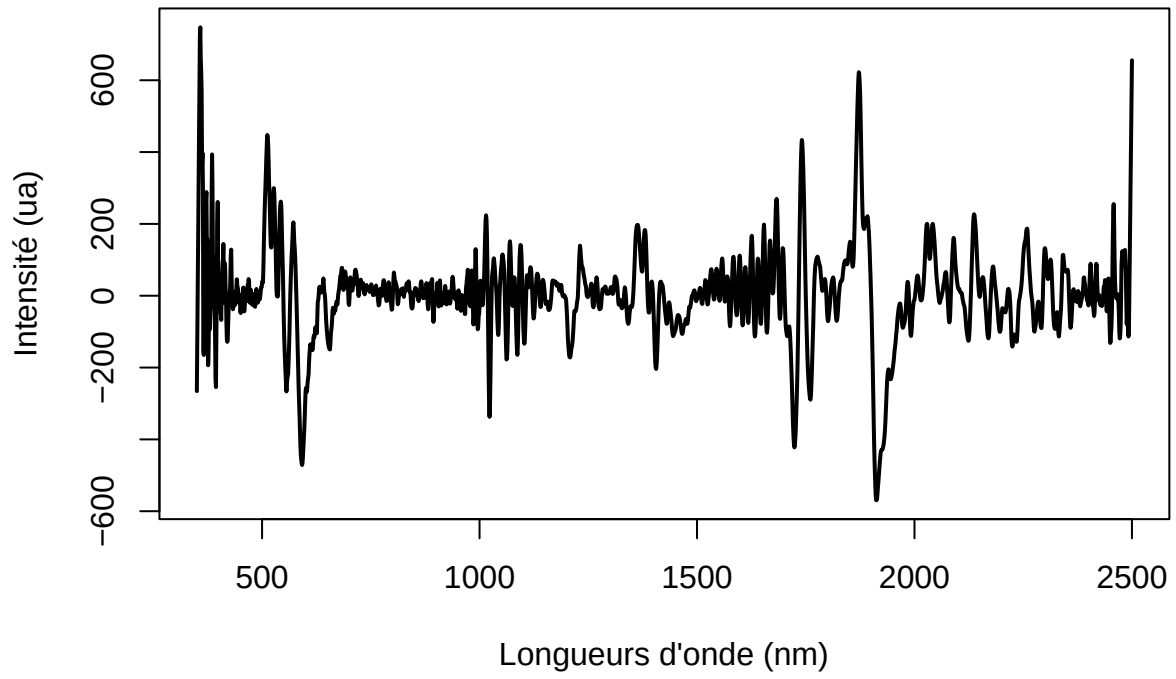
a.



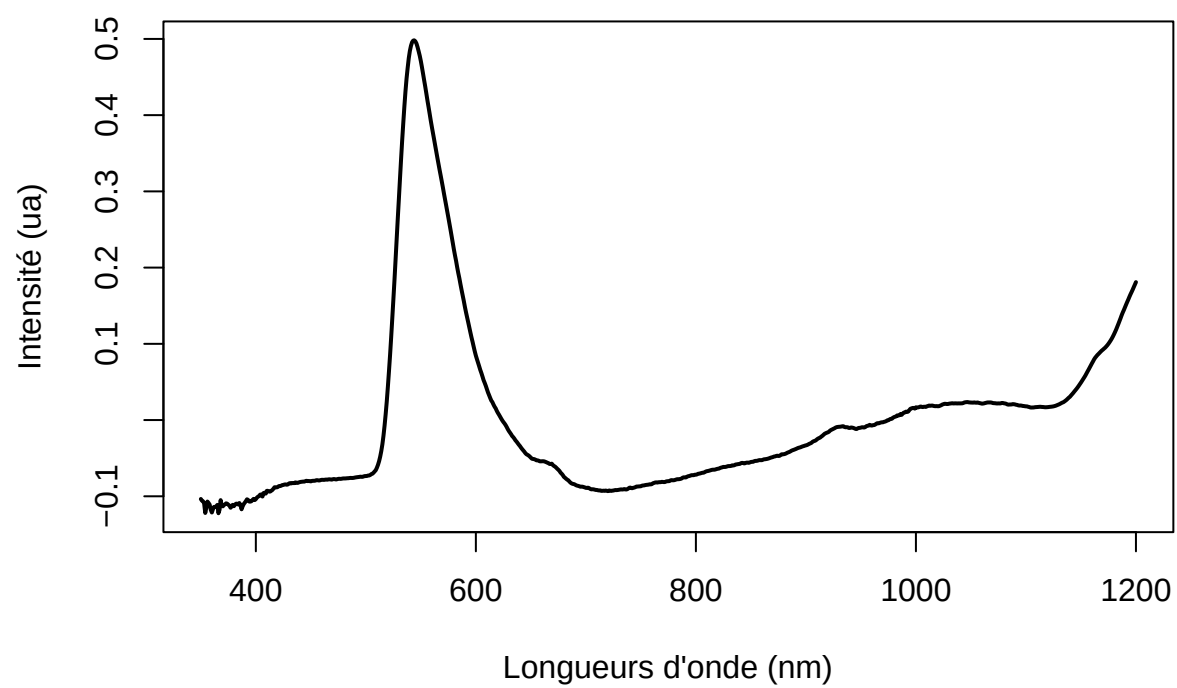
b.



C.



h



Coefficient beta pour 16:0, 18:1 et 18:2n9 avec Pré 1600-1750nm +
SNV + Der2, et 8VL

